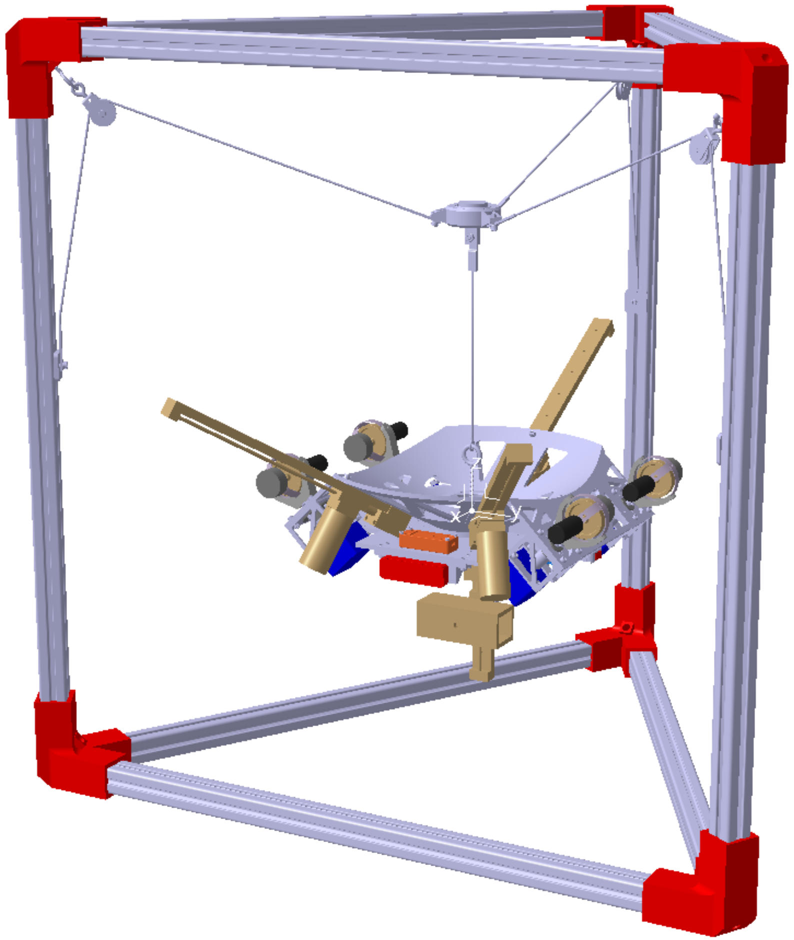
CRITICAL DESIGN REVIEW

Project ODIN

Presentation Objectives

- What Problem Does ODIN Address?
- What Objectives Must ODIN Meet?
- How Will ODIN Operate as a System of Systems?
- How is ODIN Designed as a System?
- How are ODIN's Subsystems Designed?
- How is the ODIN Program Managed?

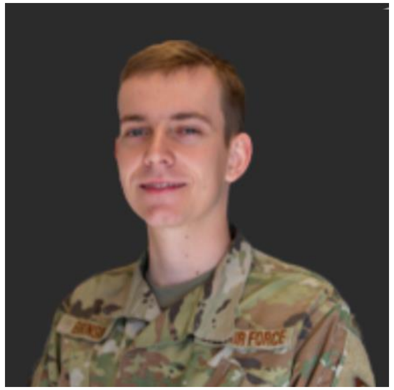


ODIN is Properly Staffed and Ready for Program Success

Shea Schmidt
Program Manager

Andrew Reynolds
Assistant Program Manager

John Anderson
Sensors Engineer

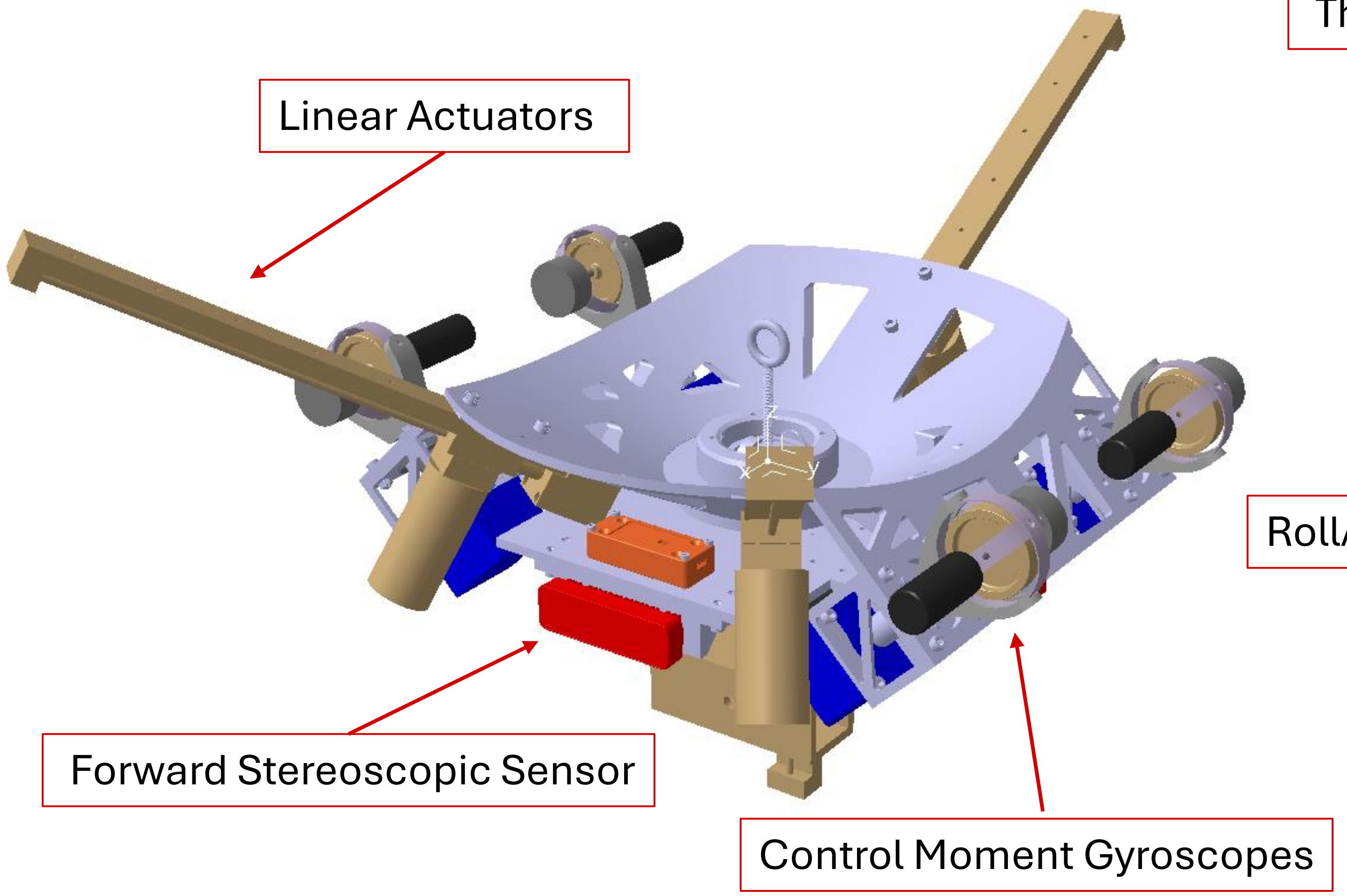


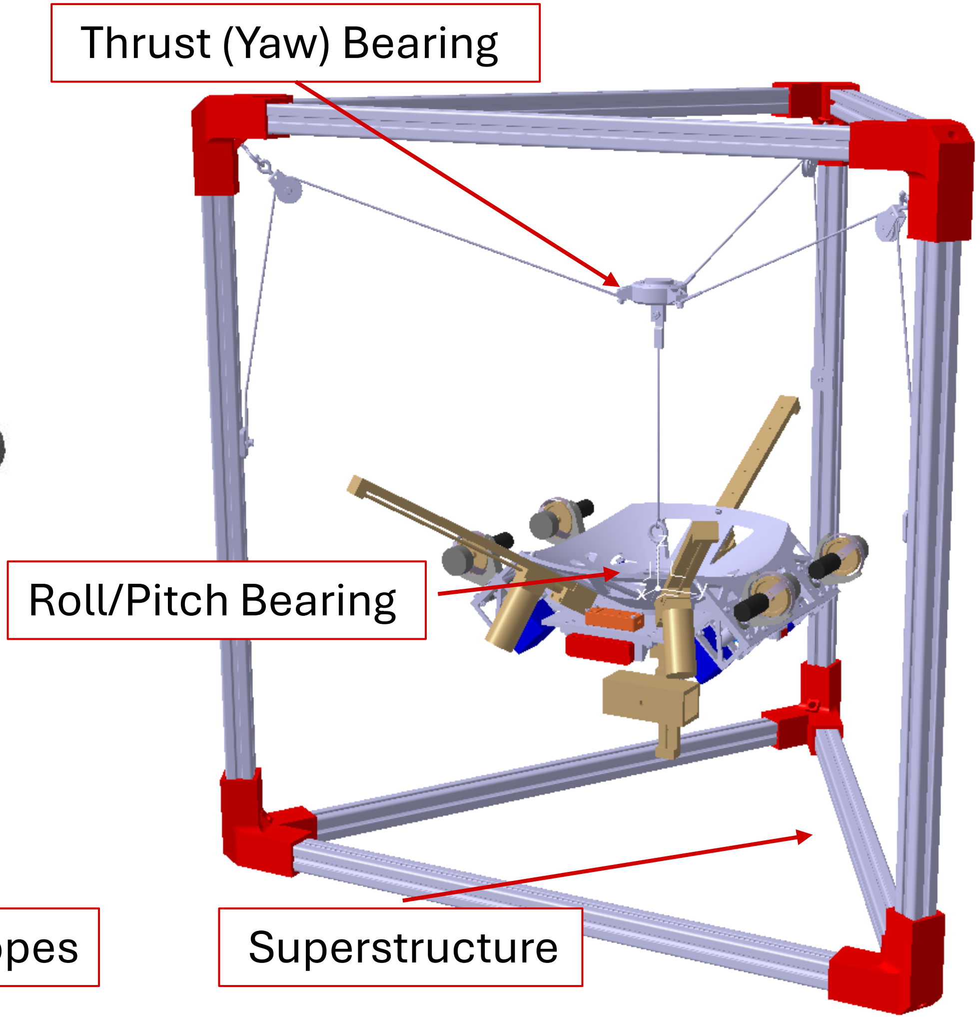
Cole Brunson
Controls Engineer

Elliot Chubon
Electrical Engineer

Adrien Hobelman
Structures Engineer

ODIN is an Advanced and Robust Controls Demonstrator

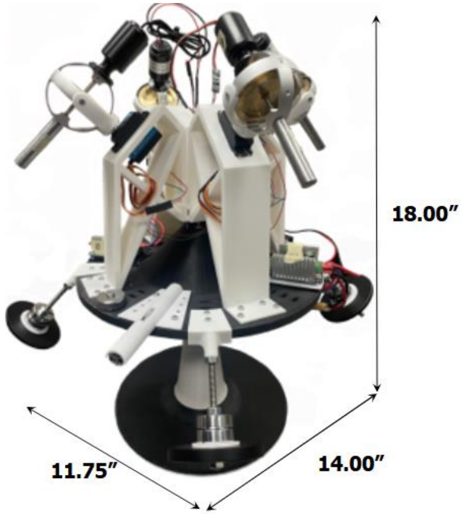


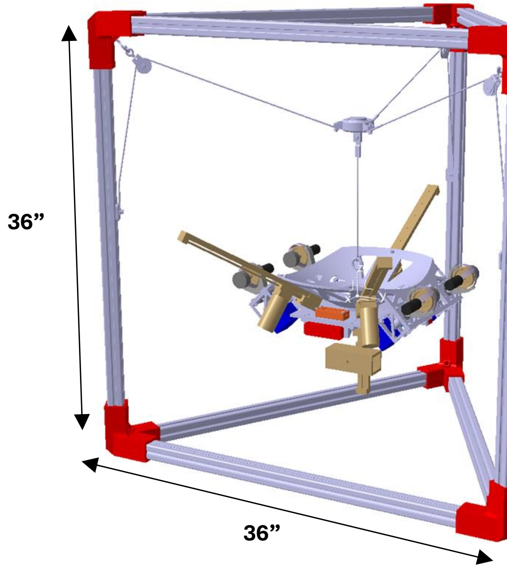


What Problem Does ODIN Address?

ODIN Improves Upon Past Controls Demonstrators

TIPSY

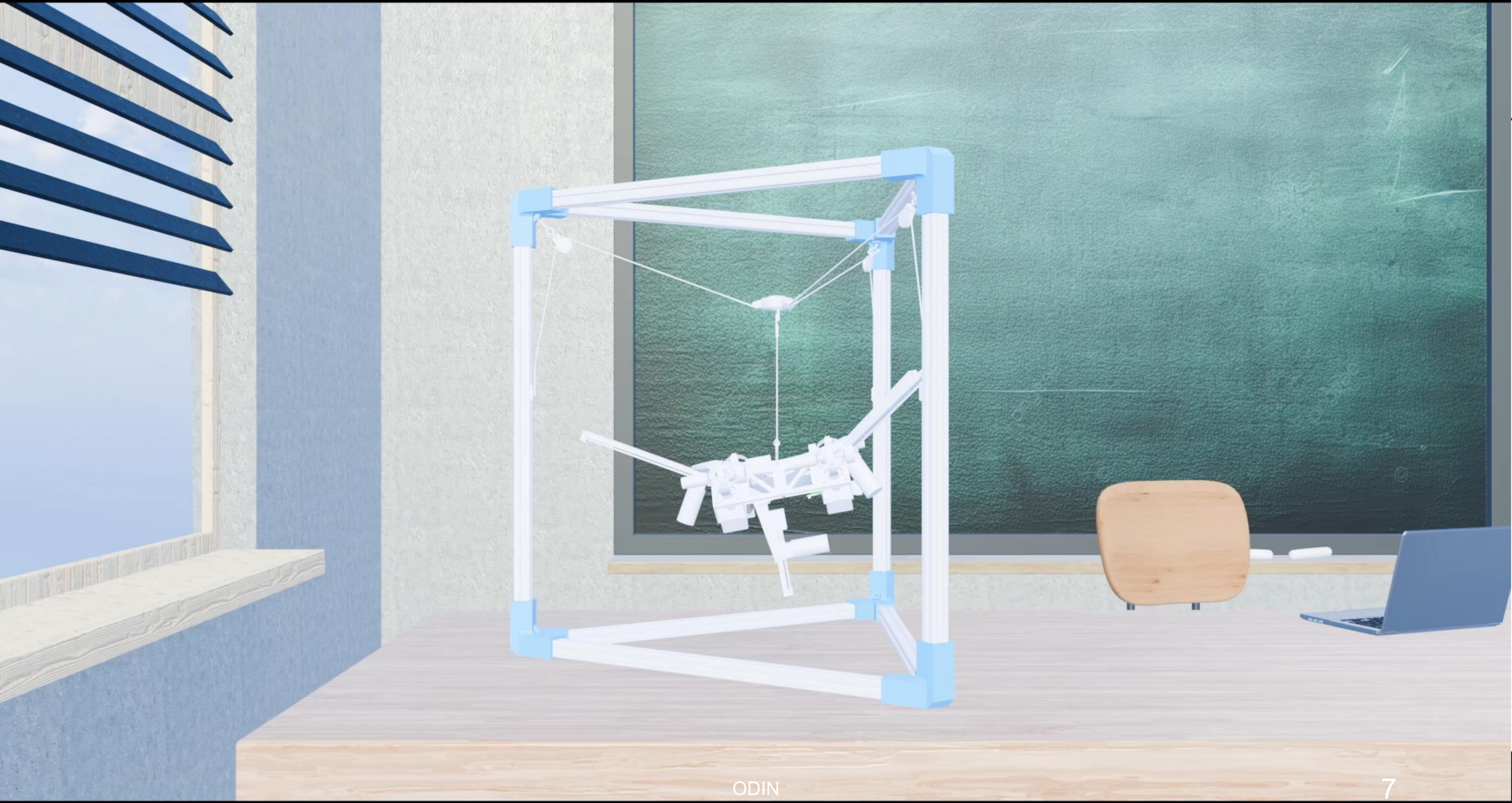




ODIN

DECISION CRITERIA	LEGACY HARDWARE	ODIN
Ease of Operation		
CMG Array Architecture		
Manufacturing Complexity		
Structural Fidelity		
Range of Motion		
Tracking Performance		
Data Handling		

No change
Improved
Degraded



Team ODIN Will Act in an Ethical Manner

- In accordance with the National Society for Engineers, Project ODIN shall maintain the following four pillars:
  - Ensure safety in all developmental and operational applications
  - Act with integrity within scope of expertise
  - Act for the customer in good faith
  - Ensure environmental, cultural, and societal impacts are well considered

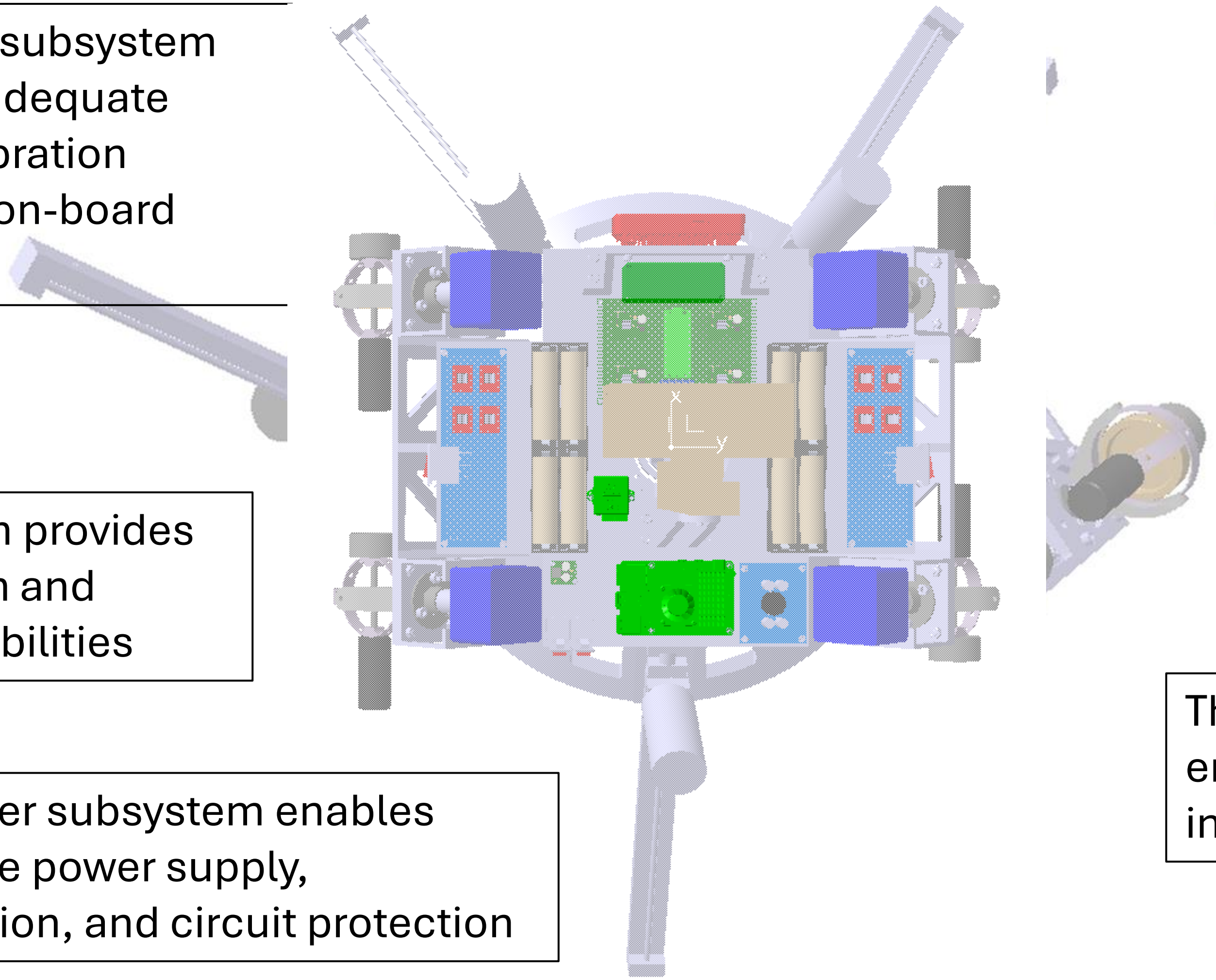
How Will ODIN Operate as a System of Systems?

System Overview

Five Subsystems Enable Minimum Viable Operation as Defined by System Objectives

The structures subsystem allows for the adequate support and vibration attenuation of on-board equipment

The controls subsystem enables the calibration of mass properties data as well as actuating precise maneuvers



The sensor subsystem provides ODIN computer vision and optical targeting capabilities

The command subsystem enables sensor fusion, software integration, and hardware control

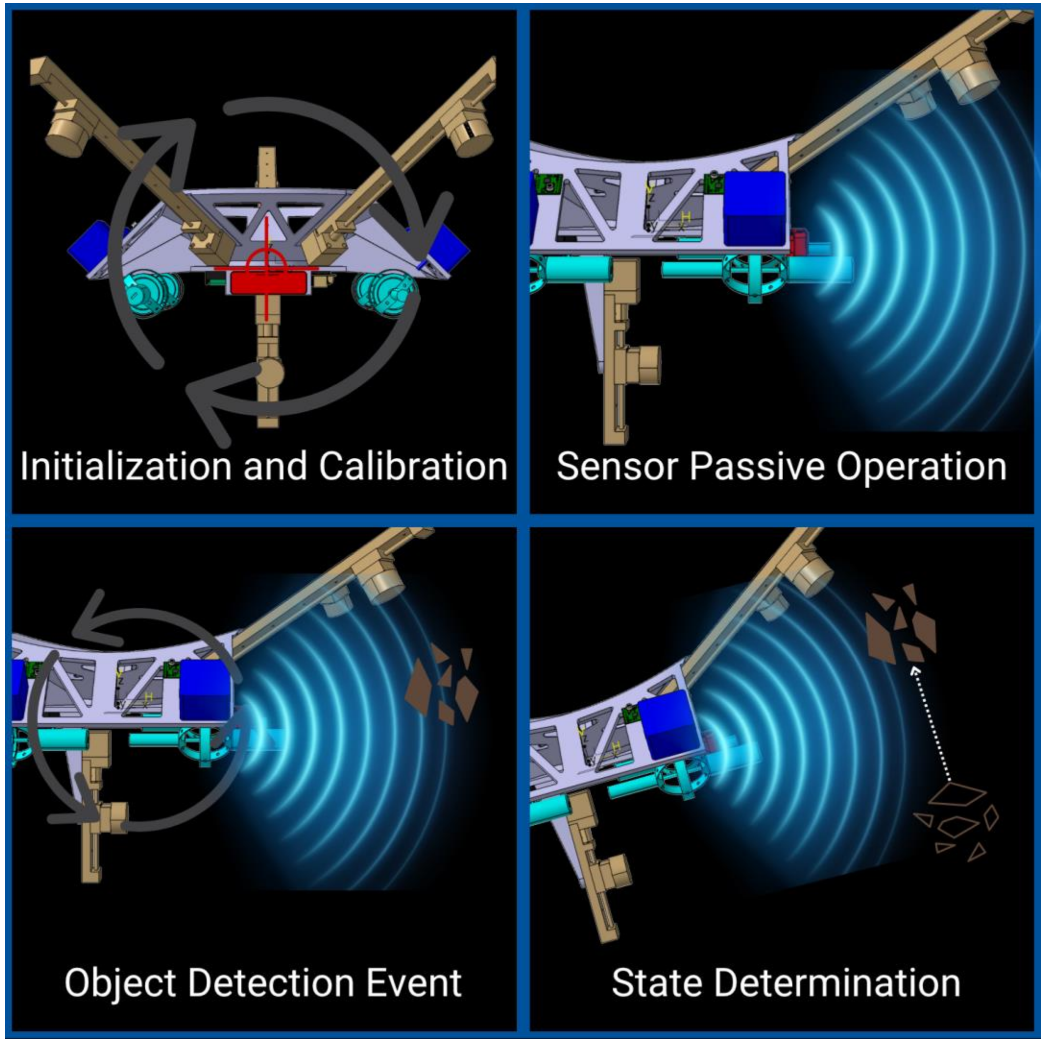
The power subsystem enables adequate power supply, distribution, and circuit protection

What Objectives Must ODIN Meet?

Concept of Operations

Customer Defined Objectives

ODIN Will Operate in Four Main Phases



Customer Defined Objectives Ensure ODIN Improves on Previous Solutions

Req ID	Title	Requirement
OBJ.2	Auto Balancing	The system should calibrate mass distribution properties.
OBJ.3	Transportability	The system should be transportable by no more than two people.
OBJ.4	Onboard Data Storage	The system should record critical data onboard for diagnostics.
OBJ.6	Debris Tracking	The system should be capable of tracking nearby objects.
OBJ.7	Independent Power	The system should function independently from external power.
OBJ.8	State Data Determination	The system should determine target state information.

How is ODIN Designed as a System?

System Level Requirements

Verification Methodologies

All System Level Requirements Derived from Objectives

Legend

↗ DeriveReqt

1 System Requirements		OBJ.1 Budget	OBJ.2 Auto balancing	OBJ.3 Transportability	OBJ.4 Data Storage	OBJ.5 Data Transmissic	OBJ.6 Tracking	OBJ.7 Independent Pow	OBJ.8 State Data	OBJ.9 Positioning	OBJ.10 Vibration Contro
		1	2	1	1	1	5	1	2	2	2
SYS.1 Budget	1	↗									
SYS.2 Vibration	1										↗
SYS.3 Roof Array	1						↗				
SYS.4 Debris Range	1						↗				
SYS.5 Battery Power	1							↗			
SYS.6 Data Interface	1					↗					
SYS.7 Mass Balancing	1		↗								
SYS.8 Superstructure	2		↗								↗
SYS.9 Cart Constraint	1			↗							
SYS.10 Debris Tracking	2						↗		↗		
SYS.11 Debris Velocity	2						↗		↗		
SYS.12 Sensor Proximity	2						↗			↗	
SYS.13 Critical Data Storage	1				↗						
SYS.14 Attitude Determination	1									↗	

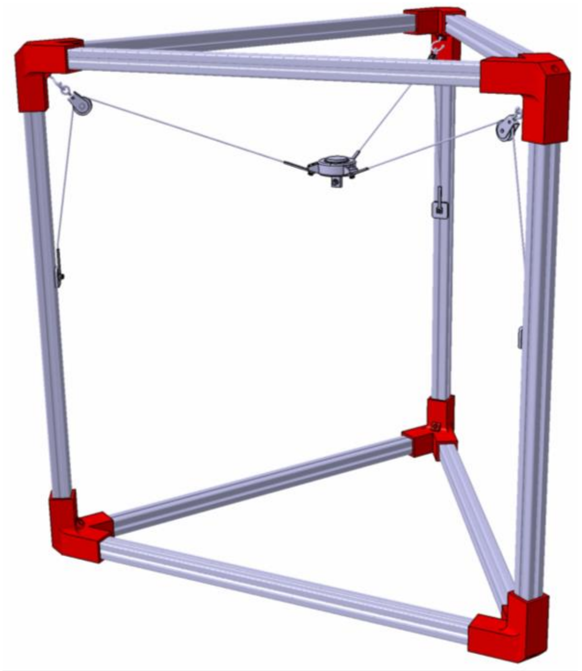
Needs Repo [1 Stakeholder Needs]

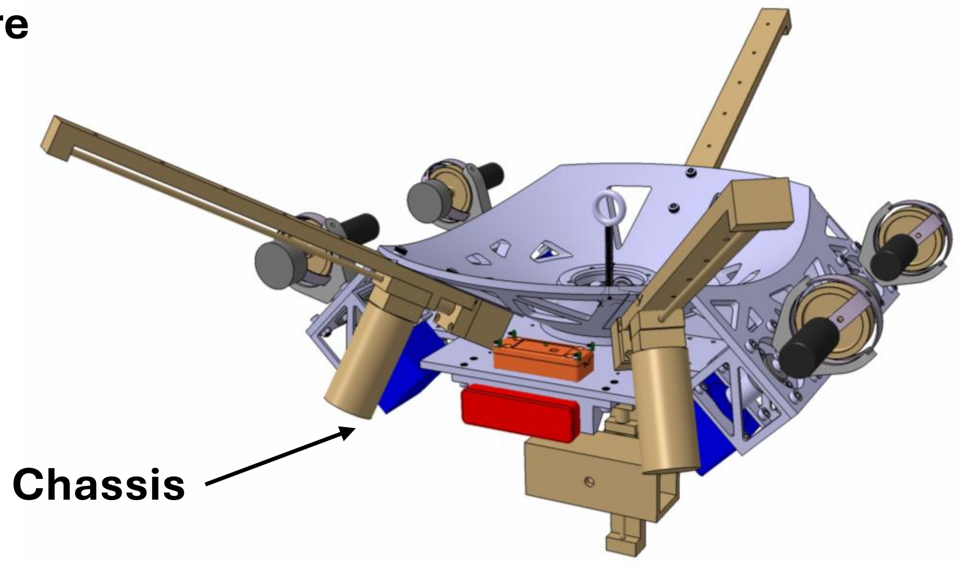
SysML Derived Requirements Matrix: System Requirements Derived From Objectives

System Level Requirements Determine Design

Req ID	Description	Verification Method	Satisfies
SYS.2	The system shall be capable of remotely sensing objects within the proximity of a sensor package.	Test	Subsystem Verified
SYS.3	The system shall have a wireless data interface for use during operation.	Demonstration	Subsystem Verified
SYS.4	The system shall be capable of determining its attitude during operation.	Demonstration	Subsystem Verified
SYS.5	The system shall record operational data to on-board storage devices.	Test	Subsystem Verified
SYS.6	The system shall be capable of determining relative target velocity.	Test	Subsystem Verified
SYS.7	The system shall be capable of determining relative target location.	Test	Subsystem Verified
SYS.8	The system shall be capable of tracking detected objects by adjusting its own attitude.	Test	Subsystem Verified
SYS.9	The system shall utilize a roof array CMG configuration.	Inspection	Subsystem Verified
SYS.10	The system shall be transportable by no more than 2 people with a maximum of 40 lbs per module.	Inspection	System Verified
SYS.11	The system shall be capable of adjusting its center of mass along three degrees of freedom.	Test	Subsystem Verified
SYS.13	The system shall incorporate and utilize independent battery power system.	Demonstration	Subsystem Verified

ODIN Utilizes A Modular Design and 3D Printed Materials to Reduce System Mass





Requirement	Requirement Value	Designed Value	Satisfies
Sys.10	<= 45 lbs per system module	44 lbs (Superstructure) 22 lbs (Chassis)	Yes

How are ODIN's Subsystems Designed?

Subsystem Level Requirements

All Control Subsystem Level Requirements Derived from System Requirements

Legend

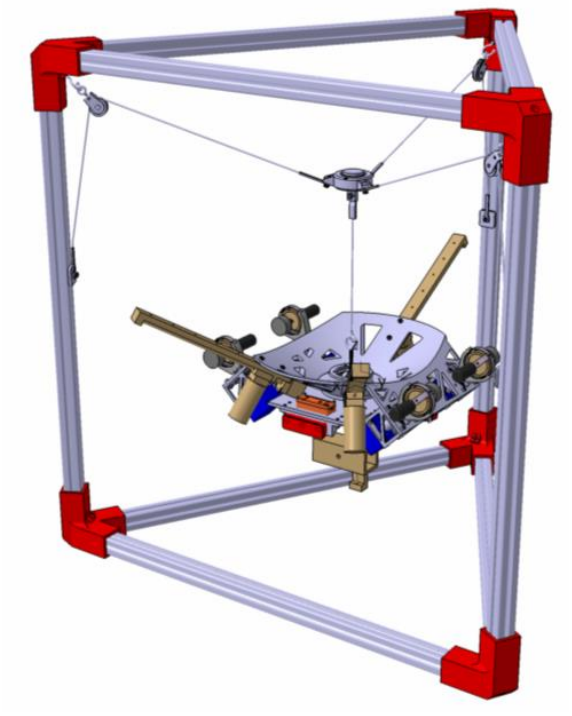
DeriveReqt

	1 System Requirements													
	SYS.1 Budget	SYS.2 Vibration	SYS.3 Roof Array	SYS.4 Debris Range	SYS.5 Battery Power	SYS.6 Data Interface	SYS.7 Mass Balancing	SYS.8 Superstructure	SYS.9 Cart Constraint	SYS.10 Debris Tracking	SYS.11 Debris Velocity	SYS.12 Sensor Proximity	SYS.13 Critical Data Storage	SYS.14 Attitude Determination
Computation Subsystem Requirements [2.2 Computation Physical Domain]					1	13	4			3		1	3	4
Control Subsystem Requirements [2.1 Control Physical Domain]						1	2			6				1
Power Subsystem Requirements [2.3 Power Physical Domain]					8	1								
Sensor Subsystem Requirements [2.4 Sensor Physical Domain]				1		1		2		3	2	4		
Structure Subsystem Requirements [2.5 Structure Physical Domain]		2	1					5	2					

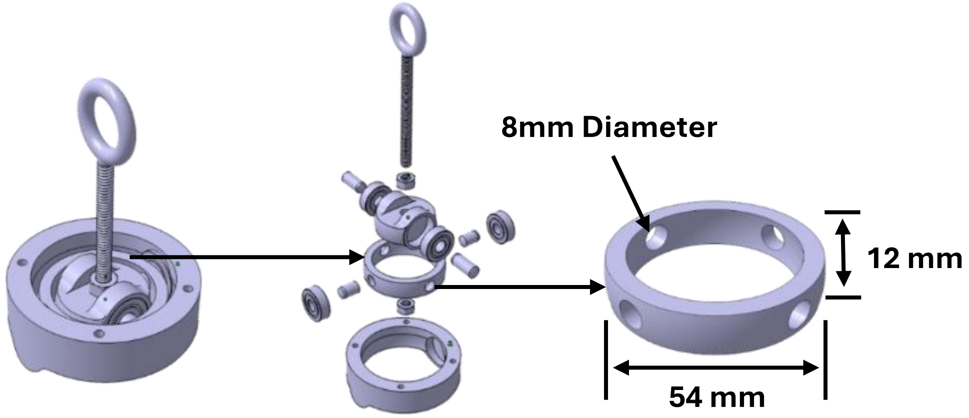
SysML Derived Requirements Matrix: Sub-System Requirements Derived From System Requirements

Structures Subsystem

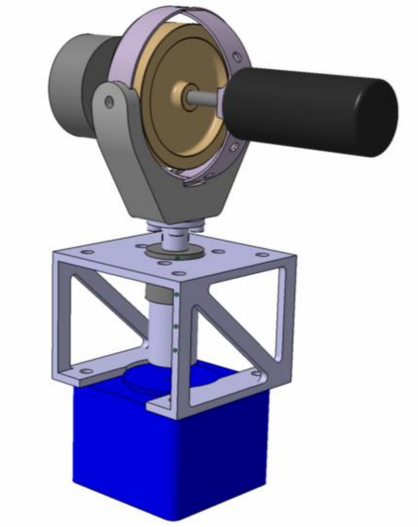
The Structures Assembly Enables Adequate Support of ODIN



Complete Assembly



Roll/Pitch Ring Assembly

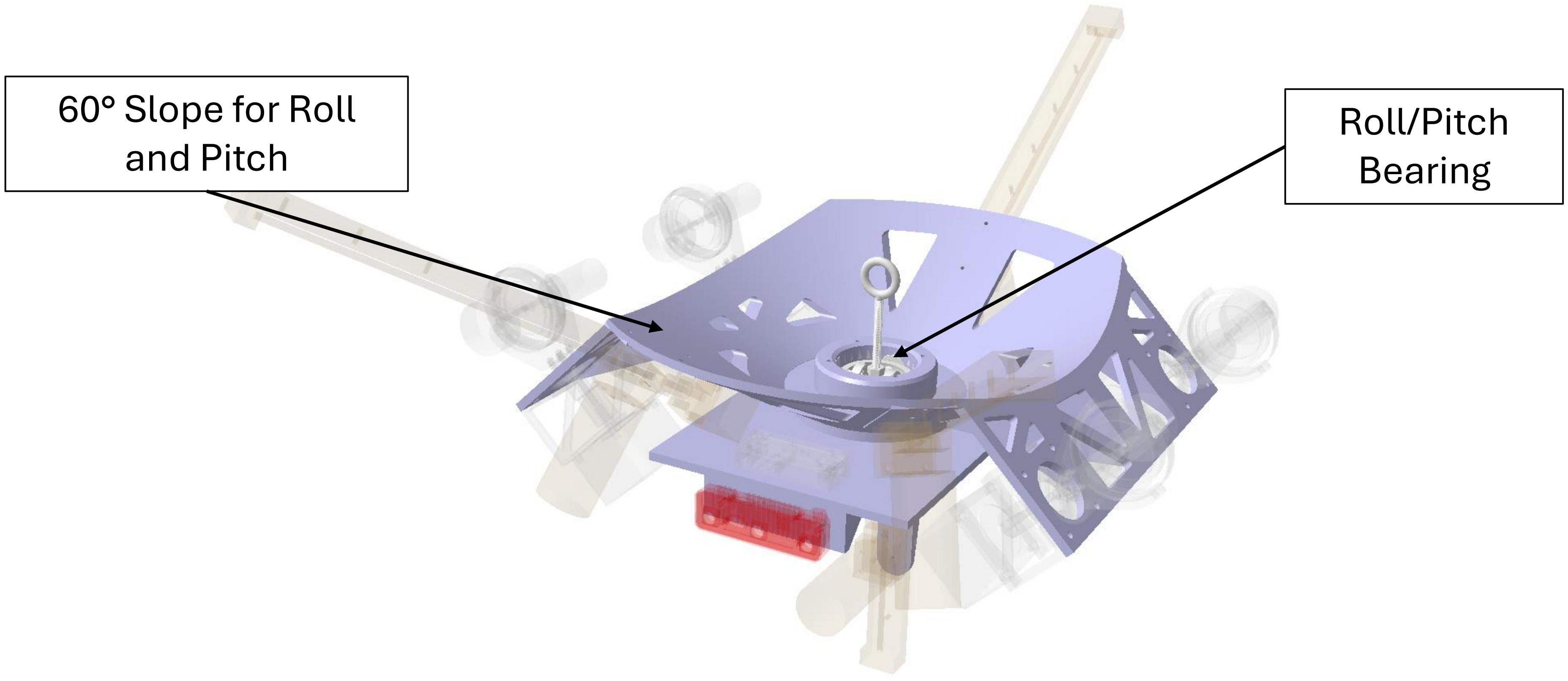


Control Moment Gyroscope Assembly

The Structures Subsystem Enables Failure Free Operation

Req ID	Description
STRCT.1	The chassis shall withstand all loads without permanent deformation or failure.
STRCT.2	The superstructure shall support the weight of the chassis without permanent deformation.
STRCT.3	The roll pitch bearing shall support the weight of the chassis without permanent deformation.
STRCT.4	The chassis shall not contact the superstructure during operation.
STRCT.5	The chassis shall be capable of rotating 55 degrees in roll and pitch.
STRCT.6	The frequency of the mode with the highest acceleration shall be less than 400Hz localized to the VN200.
STRCT.7	The frequency of the mode with the highest acceleration shall be less than 10,000Hz localized to the Slamstick.

Additively Manufactured Chassis Enables High Range of Motion

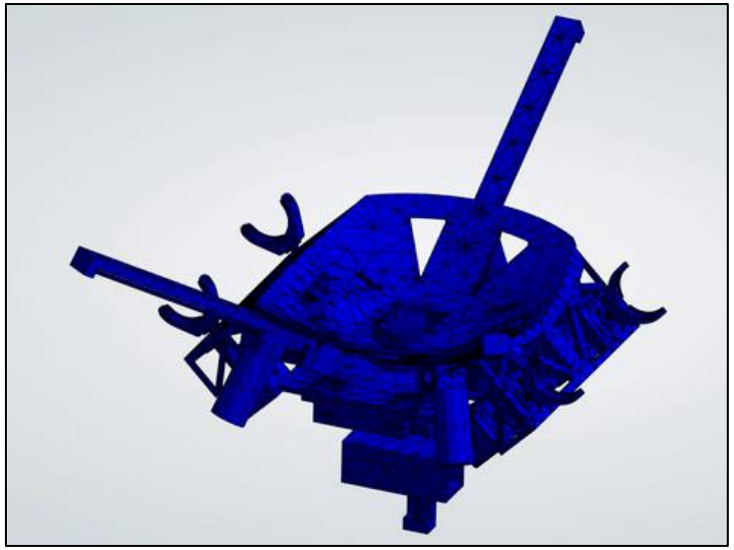


Requirement	Req Value	Analysis Value	Satisfies
Maximum Roll/Pitch Angle	≥ 55 degrees	60 degrees	Yes

The Assembly Will Maintain Structural Integrity

Max Stress in PPA-CF Roll/Pitch Ring:
9.2 MPa

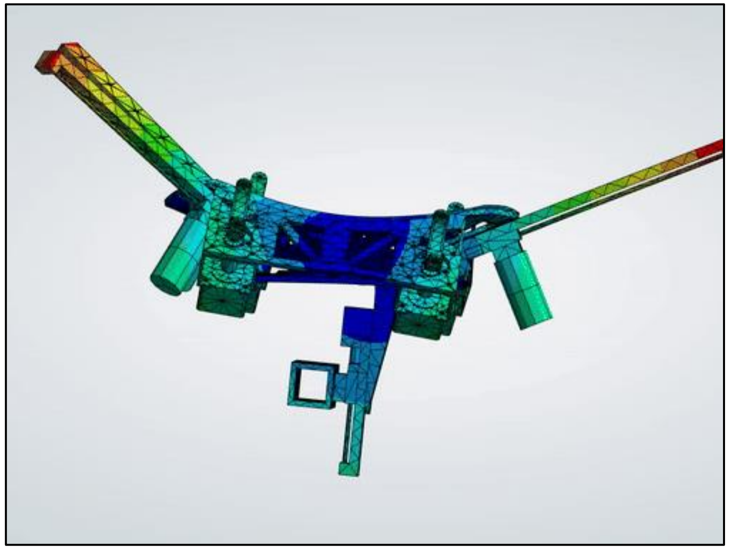
Max Stress in PETG Corner:
0.74 MPa



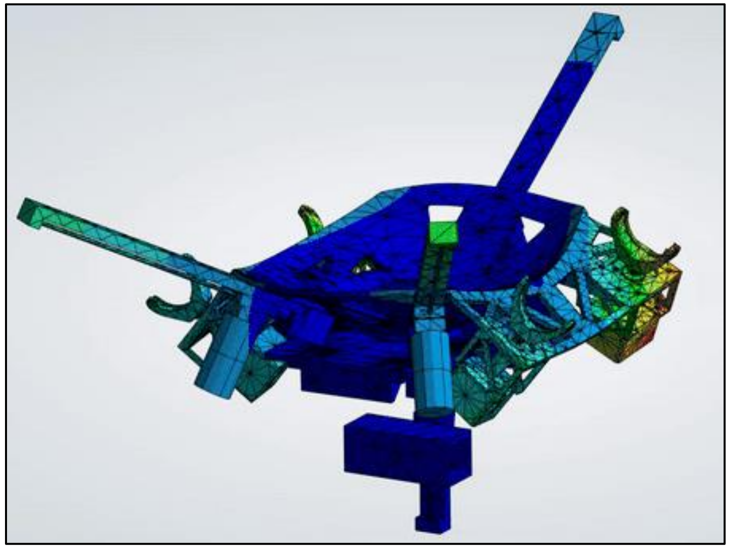
Max Stress in PETG CMG Mount:
7.2 MPa

Requirement	Req Value	Analysis Value	Satisfies
Chassis Maximum Stress	$\leq$ 23 MPa	7.2 MPa	Yes
Superstructure Maximum Stress	$\leq$ 23 MPa	0.74 MPa	Yes
Roll/Pitch Bearing Maximum Stress	$\leq$ 57 MPa	9.2 MPa	Yes

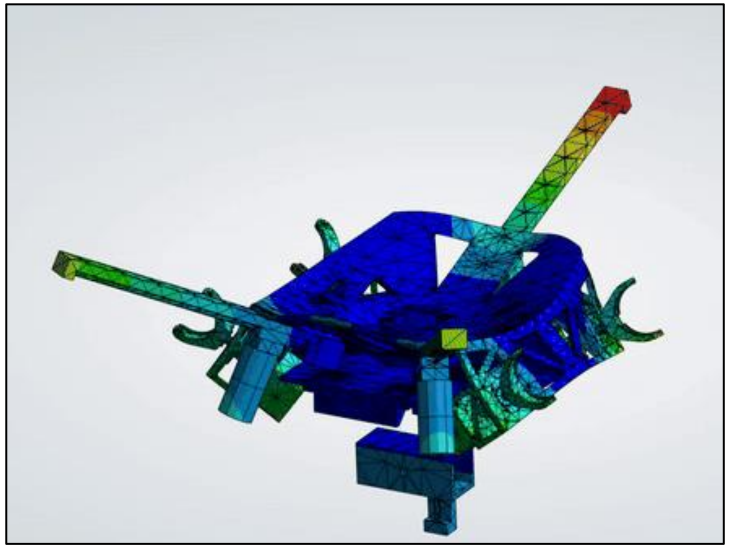
The Chassis Will Sufficiently Dampen Vibrational Loads



Max X-Axis Modal Effective Mass
Mode 3 – **26.4 Hz**



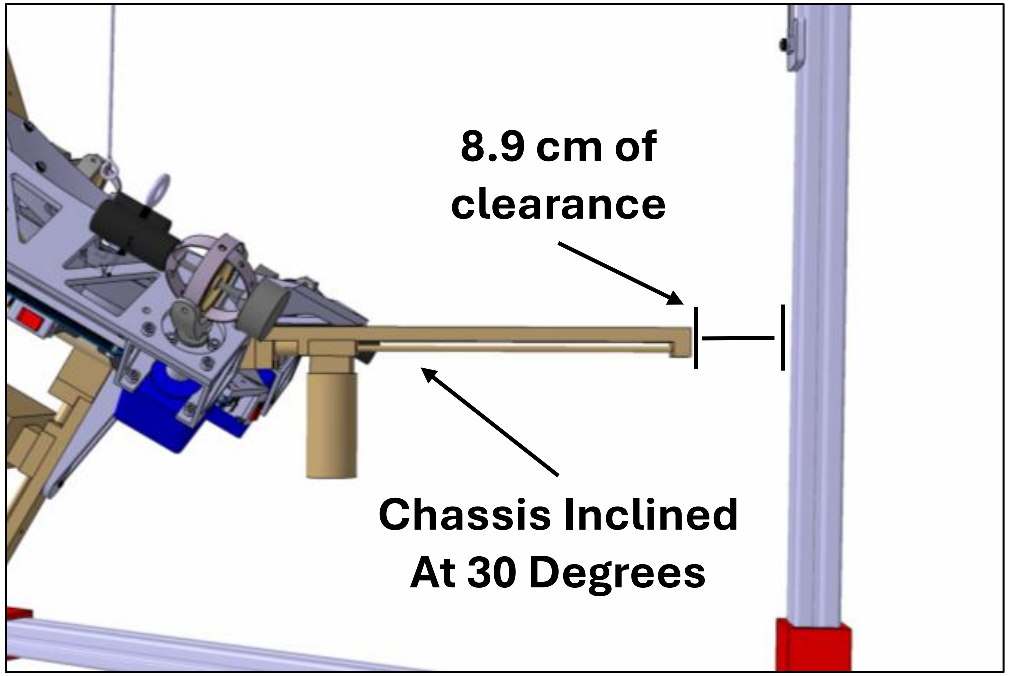
Max Y-Axis Modal Effective Mass
Mode 4 – **28.1 Hz**



Max Z-Axis Modal Effective Mass
Mode 5 – **31.6 Hz**

Requirement	Req Value	Analysis Value	Satisfies
VN200S Vibration	**≤ 400 Hz**	**26.4 – 31.6 Hz**	**Yes**
Slamstick Vibration	**≤ 10000 Hz**	**26.4 – 31.6 Hz**	**Yes**

The Chassis Will Not Contact The Superstructure



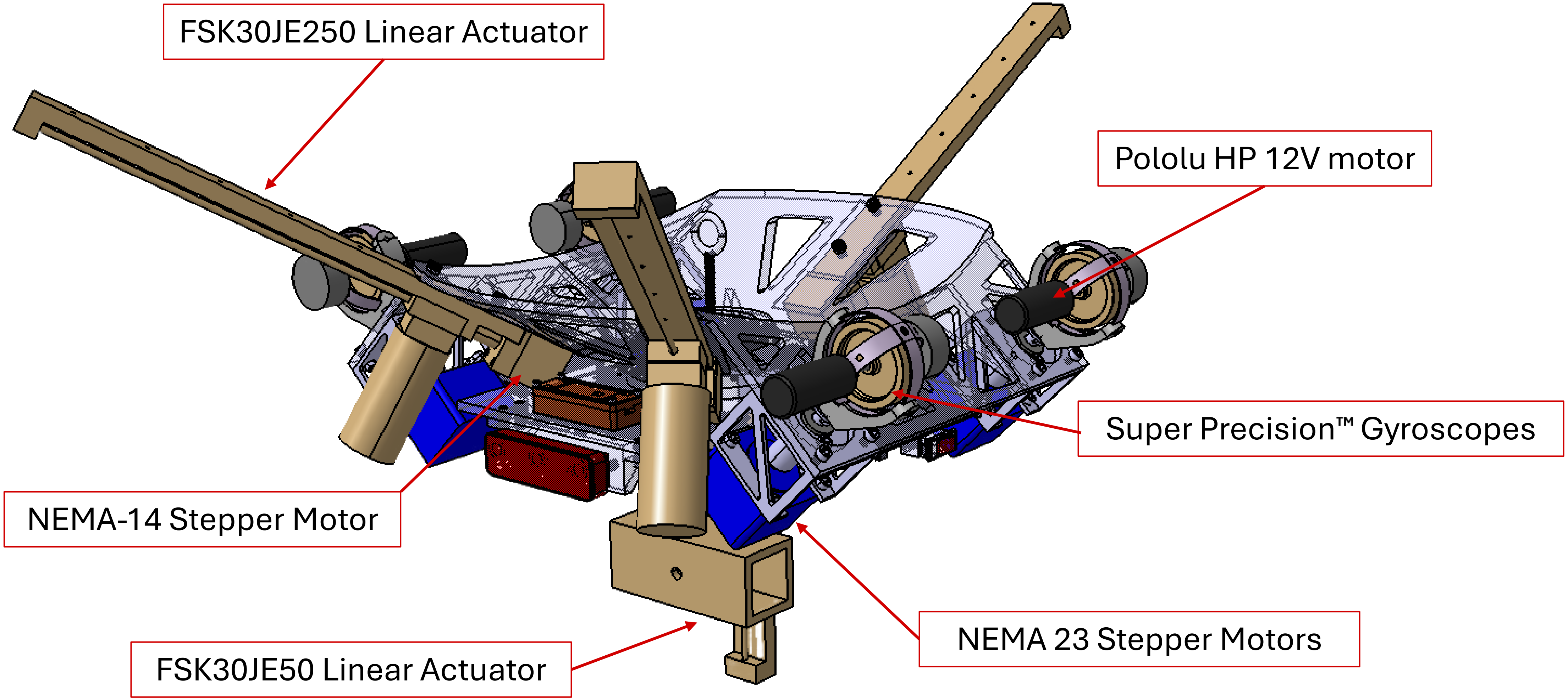
Requirement	Req Value	Analysis Value	Satisfies
Chassis Clearance To Superstructure	$\geq$ 5 cm	8.9 cm	Yes

The Structures Subsystem is Expected to Meet all Requirements

Req ID	Description	Requirement Value	Analysis Value	Satisfies
STRCT.1	Chassis Maximum Stress	$\leq$ 23 MPa	7.2 MPa	Yes
STRCT.2	Superstructure Maximum Stress	$\leq$ 23 MPa	0.74 MPa	Yes
STRCT.3	Roll/Pitch Bearing Maximum Stress	$\leq$ 57 MPa	9.2 MPa	Yes
STRCT.4	Chassis/Superstructure Clearance	$\geq$ 5 cm	8.9 cm	Yes
STRCT.5	Maximum Roll/Pitch Angle	$\geq$ 55 degrees	60 degrees	Yes
STRCT.6	VN200S Vibration	$\geq$ 400 Hz	26.4 – 31.6 Hz	Yes
STRCT.7	Slamstick Vibration	$\geq$ 10000 Hz	26.4 – 31.6 Hz	Yes

Controls Subsystem

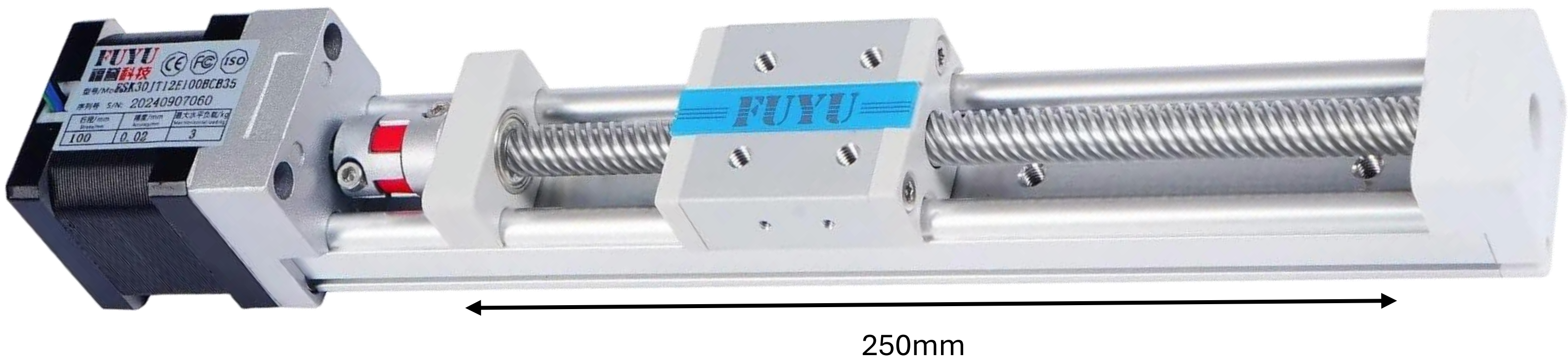
Controls Hardware Enables Mass Distribution Calibration and Attitude Control



The Controls Subsystem Enables Maneuvers and Calibration

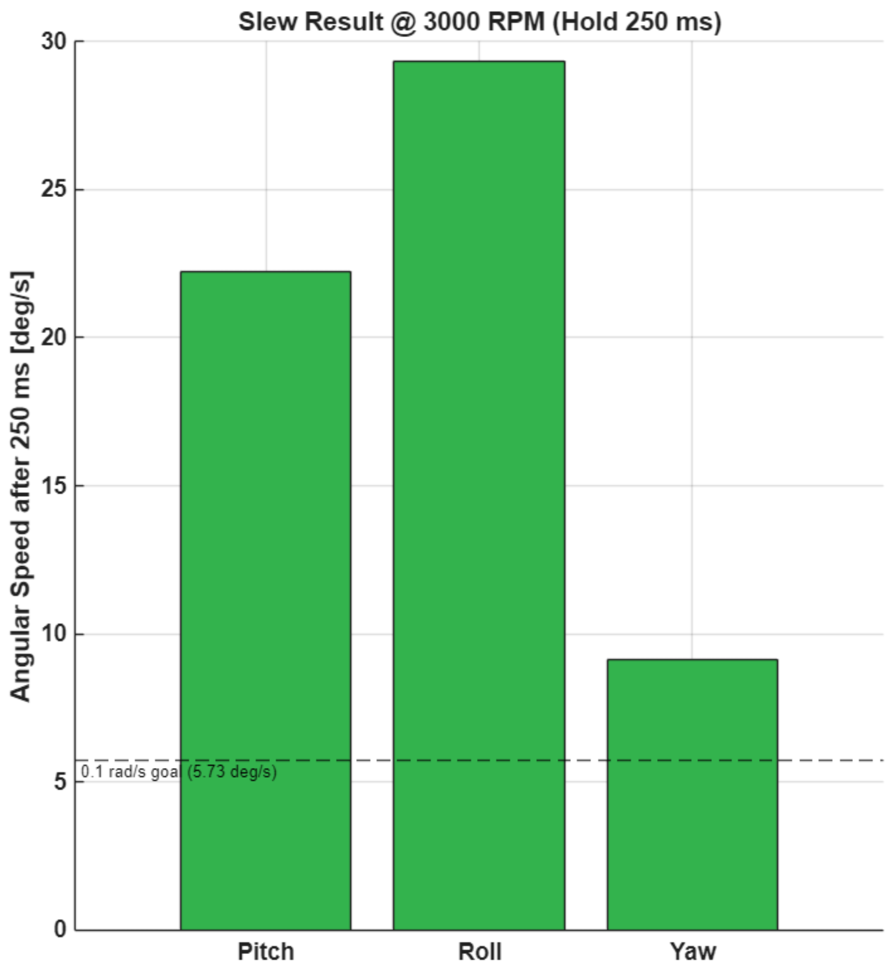
Req ID	Description
CTRL.1	The controls subsystem shall be capable of manipulating the initial center of mass (CM) to ≤ 2 mm of the anchor point along all axis.
CTRL.2-4	The controls subsystem shall command a slew rate of 5.73° (0.1 rad) per second for yaw, pitch, and roll within 250 ms.
CRTL.5	The controls subsystem shall provide at most 10% overshoot (OS) with a settling time (Ts) of 10 seconds.

Controls Hardware Can Adjust the Center of Mass to Within 2mm of Target



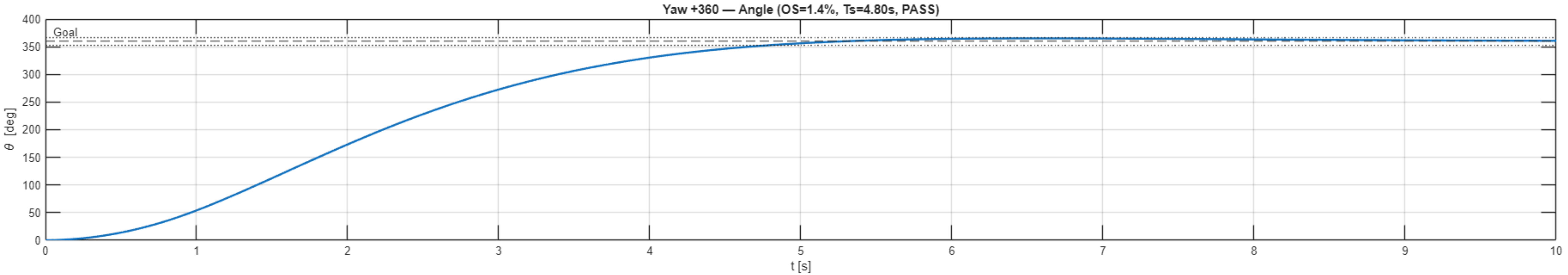
Requirement	Req Value	Analysis Value	Satisfies
CM Calibration X-Axis	≤ 2 mm	0.015 mm	Yes
CM Calibration Y-Axis	≤ 2 mm	0.017 mm	Yes
CM Calibration Z-Axis	≤ 2 mm	0.025 mm	Yes

Slew Rate Capability Ensures Rapid Target Acquisition



Requirement	Req Value	Analysis Value	Satisfies
Yaw Commanded Slew Rate	≥ 5.73° per second	9.11° per second	Yes
Pitch Commanded Slew Rate	≥ 5.73° per second	29.32° per second	Yes
Roll Commanded Slew Rate	≥ 5.73° per second	22.22° per second	Yes

Attitude Response Meets Overshoot and Settling Time Requirements



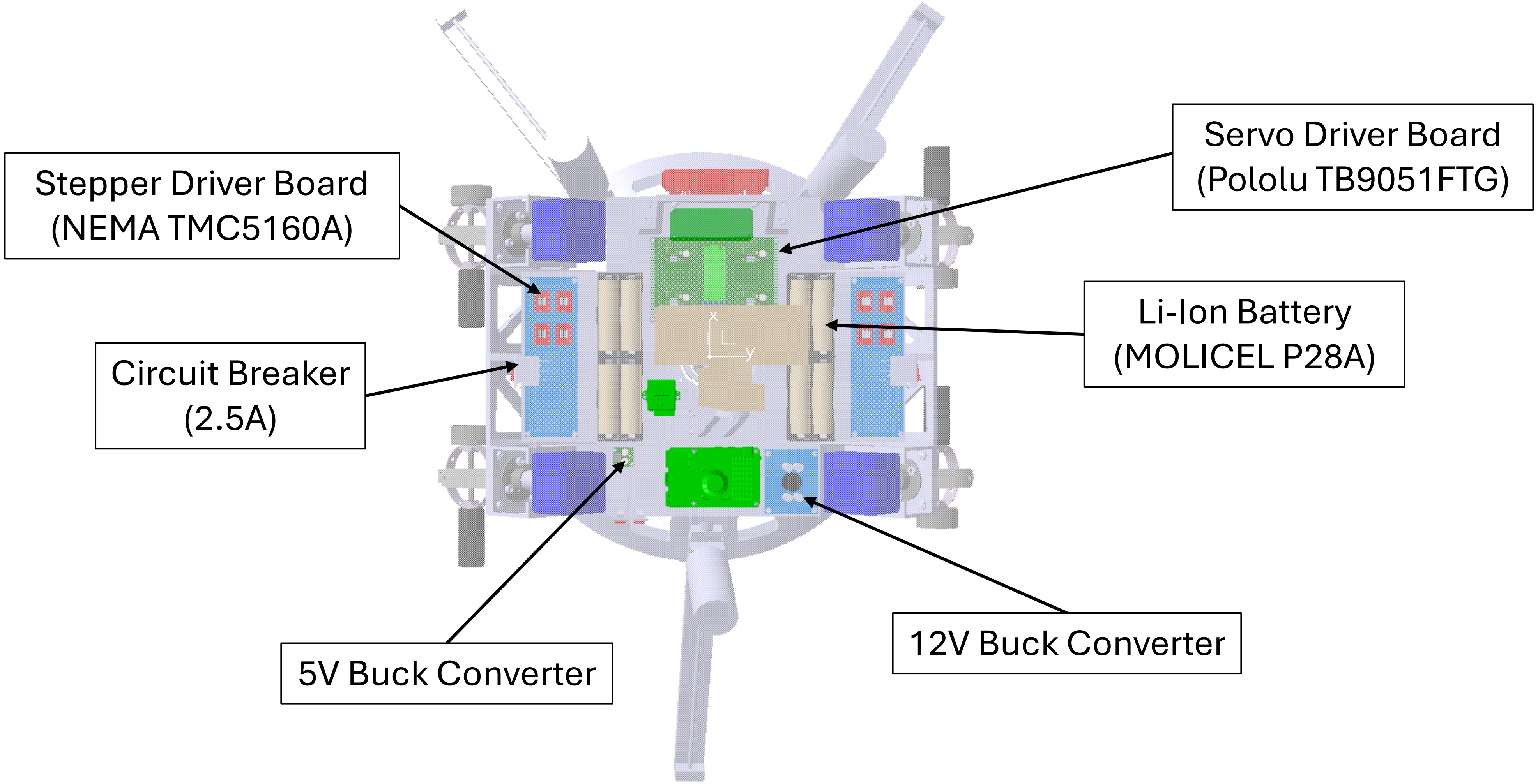
Requirement	Req Value	Analysis Value	Satisfies
Maximum Percent Overshoot	**10%**	**1.4%**	**Yes**
Maximum Settling Time	**10s**	**4.80s**	**Yes**

The Command Subsystem is Expected to Meet all Requirements

Req ID	Requirement	Requirement Value	Analysis Value	Satisfies
CTRL.1	CM Calibration	≤ 2 mm	0.025 mm	Yes
CTRL.2	Commanded slew rates	≥ 5.73° per second	9.11° per second	Yes
CTRL.3	Commanded slew rates	≥ 5.73° per second	29.32° per second	Yes
CTRL.4	Commanded slew rates	≥ 5.73° per second	22.22° per second	Yes
CRTL.5	Maximum	10% OS, 10s Ts	1.4% OS, 4.80s Ts	Yes

Power Subsystem

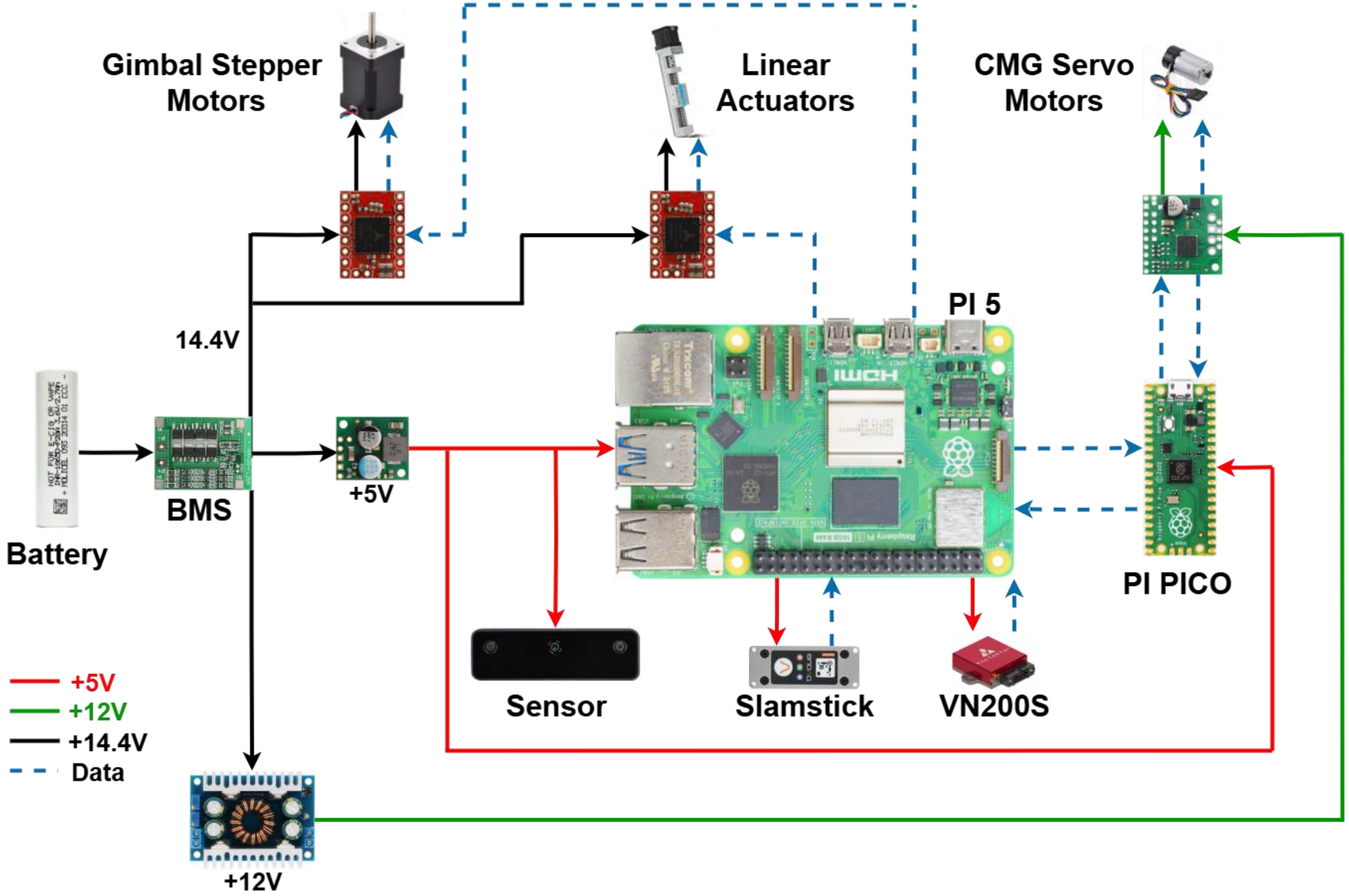
Power Hardware Enables Safe and Controlled Operation of ODIN



Power Subsystem Ensures Proper Power Delivery for ODIN

Number	Description
PWR.2	The power subsystem shall distribute required voltage to each subsystem.
PWR.3	The power subsystem shall distribute required current to each subsystem.
PWR.4	The power subsystem shall be capable of providing operational power continuously for at least 2 hours.
PWR.6	The power subsystem shall provide circuit protection to the computer, motor controllers, and motors.

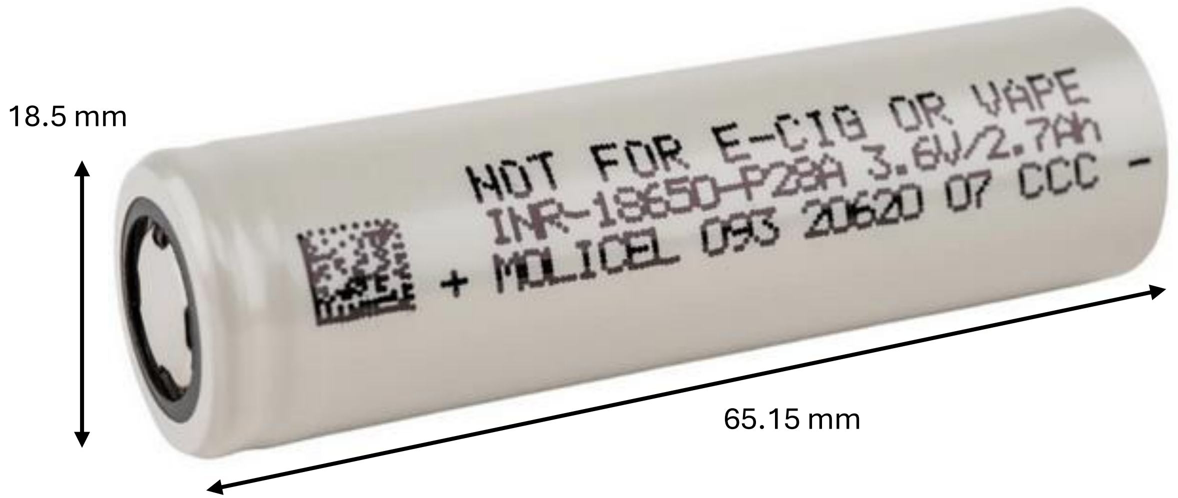
Power is Distributed in a Network to Adequately Energize Each Device



Batteries Meet ODIN Power Delivery Requirements

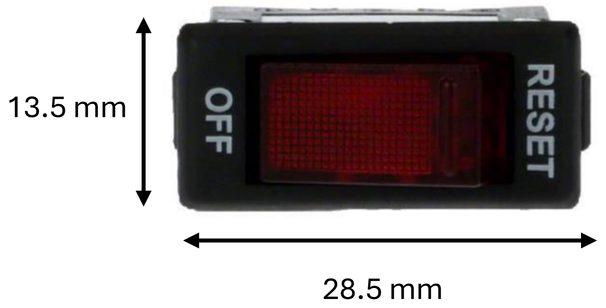
Molicel P28A Battery

- 2.8 Ah
- 3.6V Nominal per Cell
- 35A Max Discharge Rate
- 4S 2P Configuration

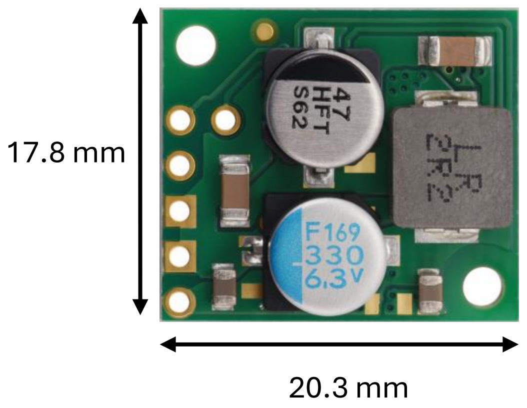


Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
Voltage	≥12V	14.4 V	Yes
Current	±10%	±8.67%	Yes
Run time	≥2 hrs	2.460 hr	Yes

Power Subsystem Provides Circuit Protection to ODIN



Circuit Breakers
- Driver Board Groups (2.5A)
- Servo Board Group (6A)
- Total Circuit (10A)



Voltage Regulators
- Convert 14.4V supply to required voltage

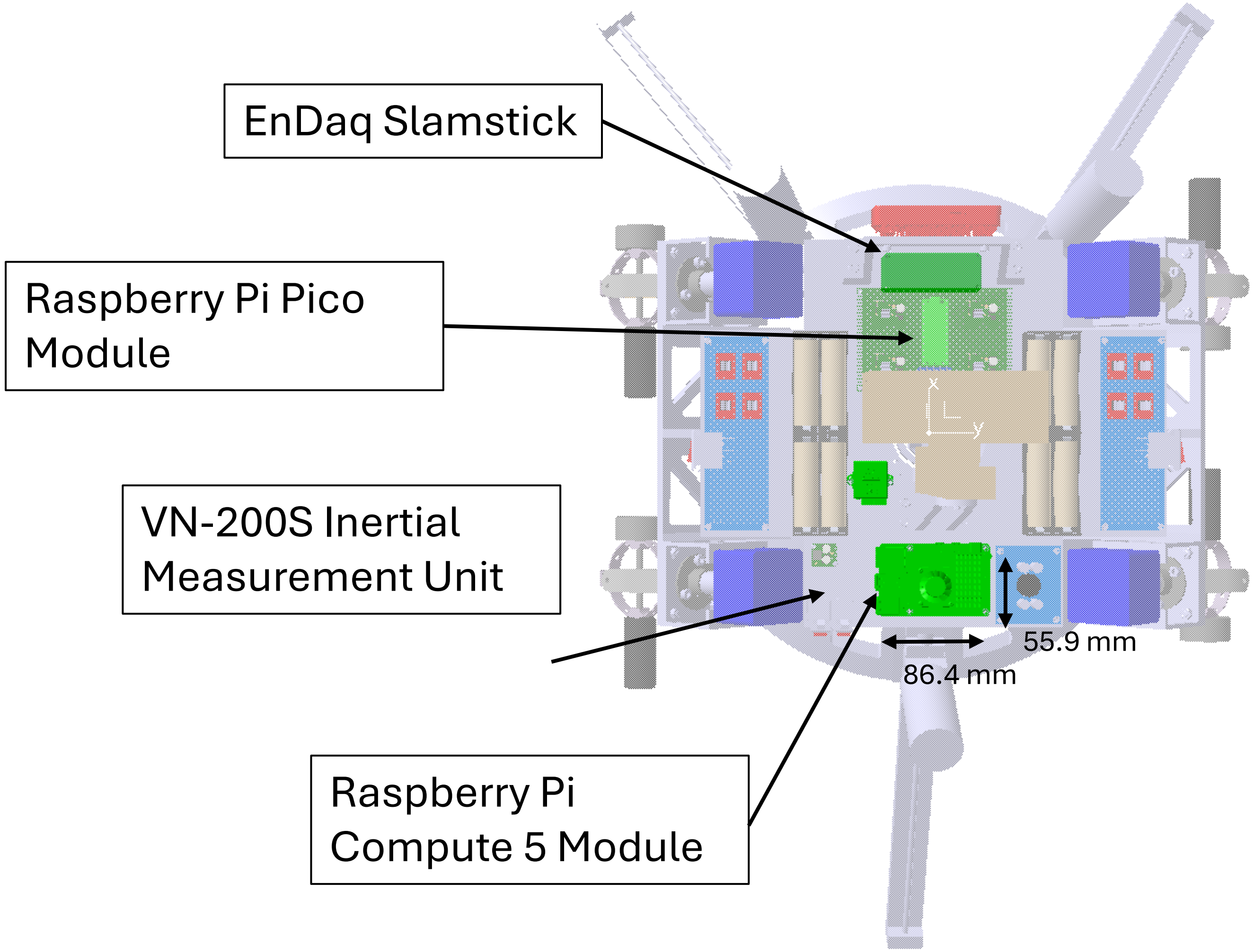
Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
Protection	FOS ≥200%	FOS 600%	Yes

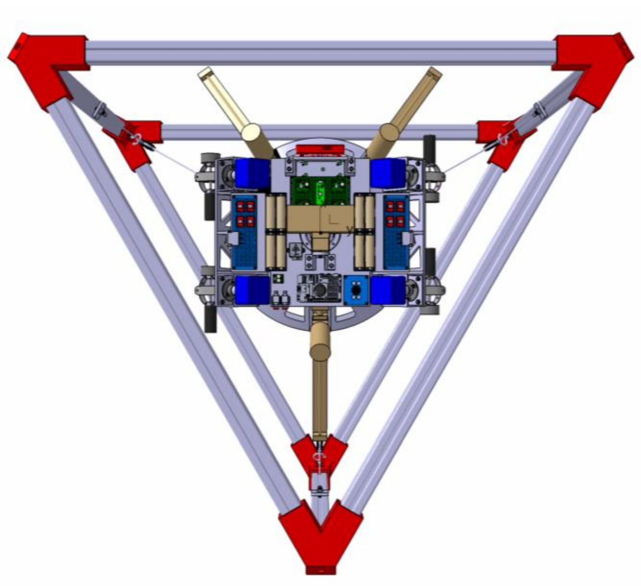
Power Subsystem is Expected to Meet all Requirements

Req ID	Description	Required Value	Analysis Value	Satisfied
PWR.2	Voltage Distribution	≥ 12V	14.4 V	Yes
PWR.3	Current Distribution	±10%	±8.67%	Yes
PWR.4	Operational Power	2 hr	2.460 hr	Yes
PWR.6	Circuit Protection	200%	600%	Yes

Command Subsystem

Command Subsystem Allows for Smart Command and Control of ODIN

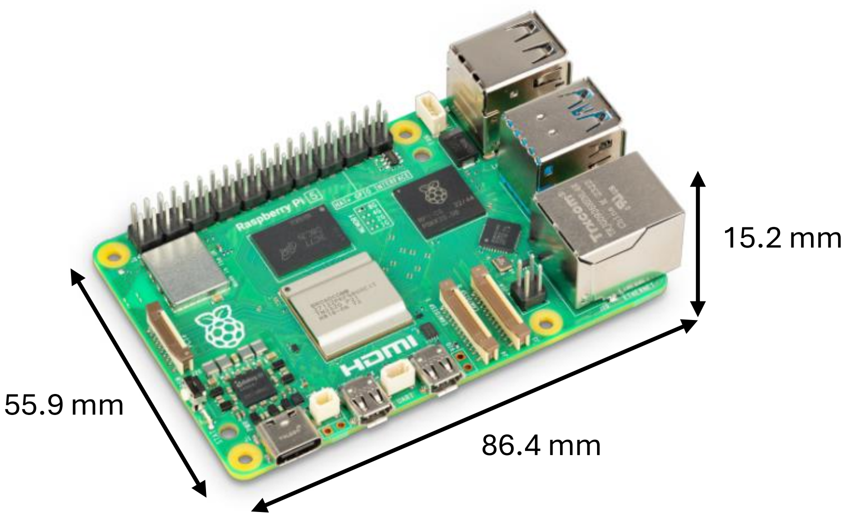




Command Subsystem Enables On-board Computation and Attitude Sensing

Req ID	Description
CMP.1-CMP.12	Subsystem shall report, record, transmit, and receive platform attitude information at a rate of at 10 Hz.
CMP.13	Subsystem shall align center of mass location to within 1mm of chassis anchor.
CMP.14	The computation subsystem shall apply corrections to calculated primary axis moments of inertia to within 5% of measured.
CMP.15	The computation subsystem shall prohibit induced oscillations of the platform within 2 seconds.
CMP.18	The computation subsystem shall make control law descisions at a frequency of at least 100 Hz.
CMP.19	The computation subsystem shall provide real-time system health monitoring capable of system shutdown in at most 100 ms after a fault is detected.

The Raspberry Pi Compute 5 Supports Monitoring Required On-board ODIN (CMP.1-CMP.12, CMP.19)



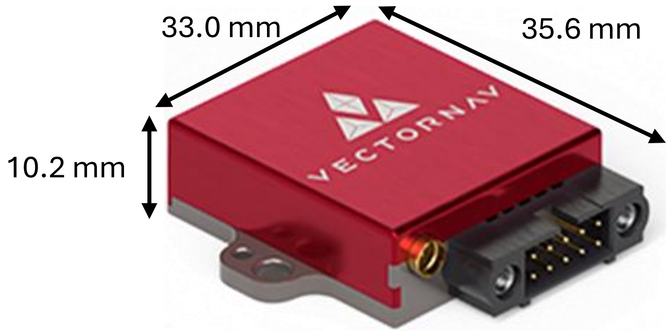
Raspberry Pi 5

Raspberry Pi 5

- 16GB SDRAM
- Dual-band Wi-Fi®
- Bluetooth 5.0/Bluetooth Low Energy (BLE)
- 40-pin header
- 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU

Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
Data Monitoring	≥10 Hz	200 Hz	Yes
Fault Monitoring	≤ 100 ms	10 ms	Yes

The VectorNav VN200S Inertial Measurement Unit Samples Attitude Fast and Accurately Enough to Control ODIN (CMP.13,14,18)



VN200S IMU

VN200S IMU

- 0.03° Dynamic pitch/roll accuracy
- 800Hz IMU sampling rate
- Accelerometer: 0.0005 g resolution
- Gyroscope: 0.02°/s resolution
- Real time Kalman filtering and compensation

Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
CM Calibration	≤ 1 mm	0.559 mm	Yes
MMOI Calibration	≤ 5% kg·m2	≤ 1% kg·m2	Yes
Control Law	≥100 Hz	800 Hz	Yes

The EnDaq Slamstick Enables Oscillation Prevention and Vibration Data Acquisition (CMP.15)

Endaq Slamstick X

Endaq Slamstick X

- Digital capacitive and piezoelectric
- 3,200-20,000 Hz sampling rate
- 0.004-0.06 g resolution
- Gyroscope: 0.06°/s resolution @ 200Hz

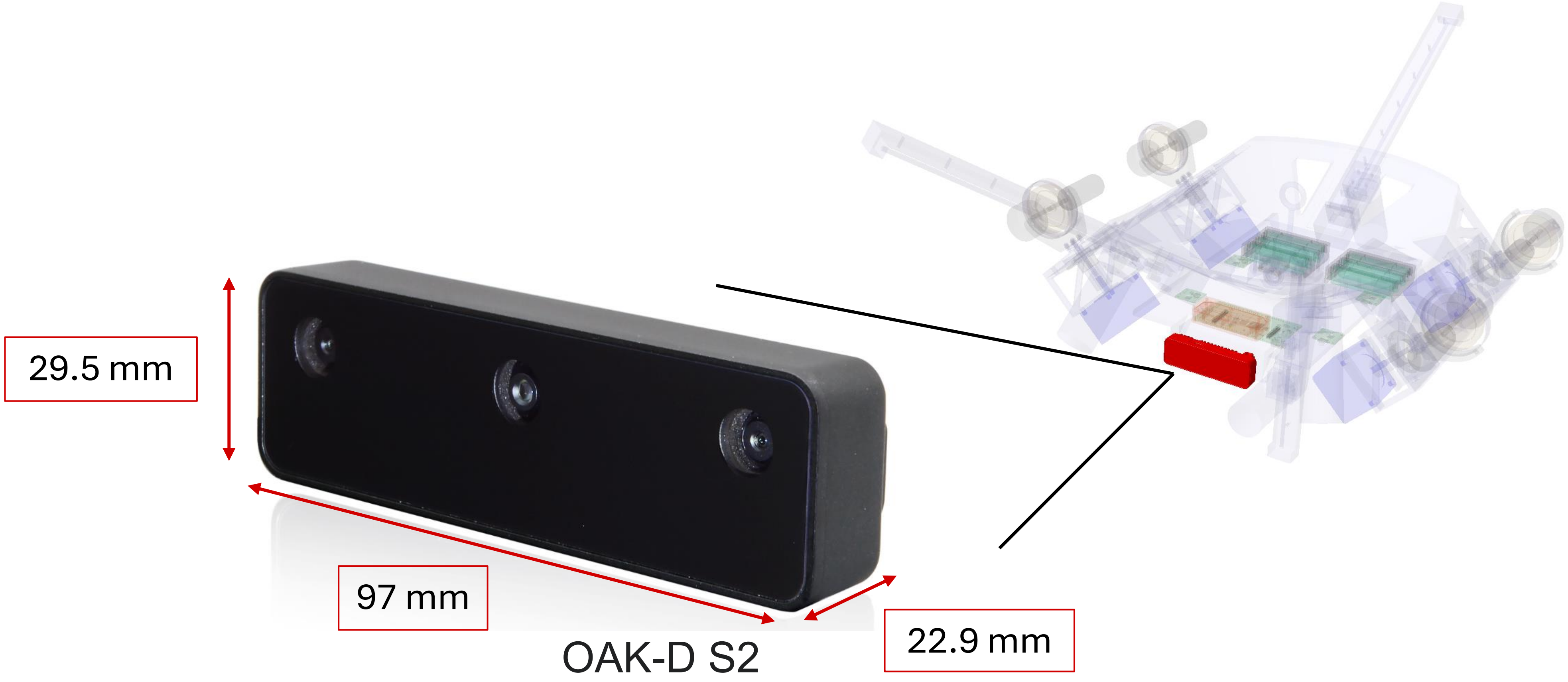
Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
Induced Oscillation	≤ 2 s	0.0025 s	Yes

The Command Subsystem is Expected to Meet all Requirements

Req ID	Requirement	Required Value	Analysis Value	Satisfied
CMP.1-CMP.12	Report, record, transmit, receive attitude date.	≥10 Hz	200 Hz	Yes
CMP.13	Center of mass calibration	≤ 1mm	0.559mm	Yes
CMP.14	Mass moment of inertia calibration	≤ 5% of analytical	≤ 1%	Yes
CMP.15	Oscillation prevention	≤ 2 s	100 ms	Yes
CMP.18	Fault monitoring	≤ 100 ms	10 ms	Yes

Sensor Subsystem

Sensor Subsystem Enables Object Tracking



Sensor Subsystem Enables Object Tracking

Number	Description
SEN.2	The sensor subsystem shall provide a minimum of 45° diagonal field of view.
SEN.3	The sensor subsystem shall provide targeting telemetry data at a minimum of 15 Hz.
SEN.4	The sensor subsystem shall determine depth up to at least 3 meters.
SEN.7	The sensor subsystem shall be capable of detecting an object with at least a 7 cm diameter.
SEN.9	The sensor subsystem shall be capable of reporting angular position within ±4° total error.

Supplier Specifications Meet Hardware Requirements

Variant	Color Camera	Stereo Pair
Sensor	IMX378 (PY052)	OV9282 (PY003)
DFOV / HFOV / VFOV	82° / 69° / 55°	89,5° / 80° / 55°
Depth	N/A	.70 – 12 m

Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
DFOV	**≥45°**	**82°**	**Y**
Depth	**≥3m**	**12m**	**Y**

Source: https://docs.luxonis.com/hardware/products/OAK-D%20S2

Supplier Benchmarking Meets Frequency Requirement

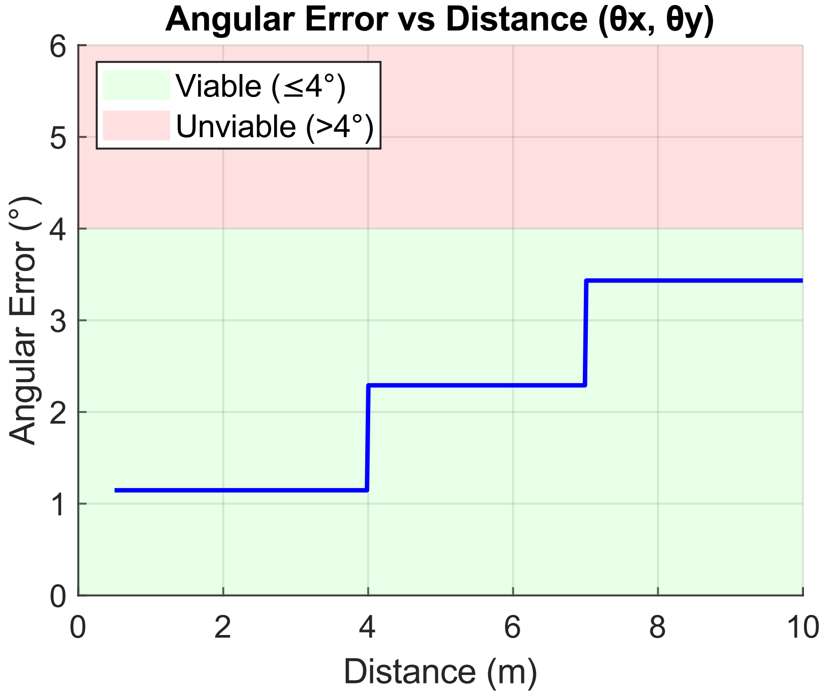
Model name	Shape	NN Latency	NN Frequency (Hz)
DeepLab V3	256 x 256	48.12 ± 0.03 ms	36.53 ± 0.64
DeepLab V3	513 x 513	253.12 ± 0.04 ms	6.32 ± 0.09
YOLOv6n R2	416x416	29.28 ± 0.02 ms	65.51 ± 0.06
YOLOv6n R2	**640x640**	**66.42 ± 0.03 ms**	**29.30 ± 0.16**
YOLOv6t R2	416x416	54.09 ± 0.05 ms	35.78 ± 0.06

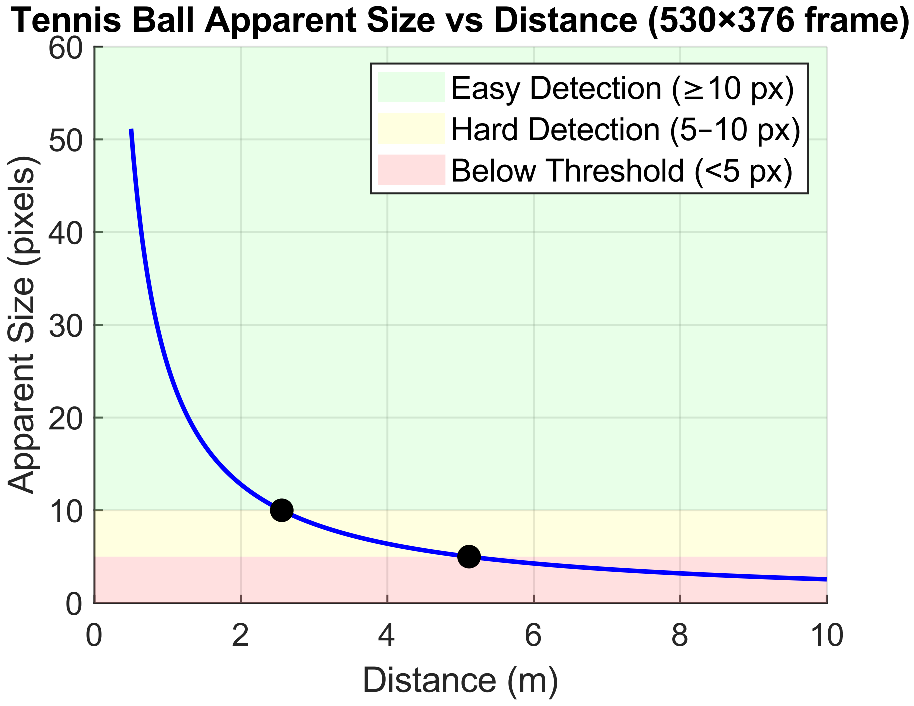
Requirement	Requirement Value	Analysis Value	Satisfies (Y/N)
Frequency	≥15 Hz	29 Hz	Y

Source: https://docs.luxonis.com/hardware/platform/rvc/rvc2

NN: Neural Network

Analysis Meets Depth and Precision Requirements





Requirement	Requirement Value	Analysis Value	Satisfies
Angular Error	≤4°	±2.5°	Y
Target Detection	≥3m	5m	Y

Sensor Subsystem is Expected to Meet all Requirements

Req ID	Requirement	Required Value	Analysis Value	Satisfied
SEN.2	Diagonal FOV	≥45°	82°	Yes
SEN.3	Data Frequency	≥15 Hz	29 Hz	Yes
SEN.4	Depth Range	≥3m	12m	Yes
SEN.7	Target Detection	≥3m	5m	Yes
SEN.9	Angular Error	≤4°	±2.5°	Yes

How is ODIN as a Program Managed?

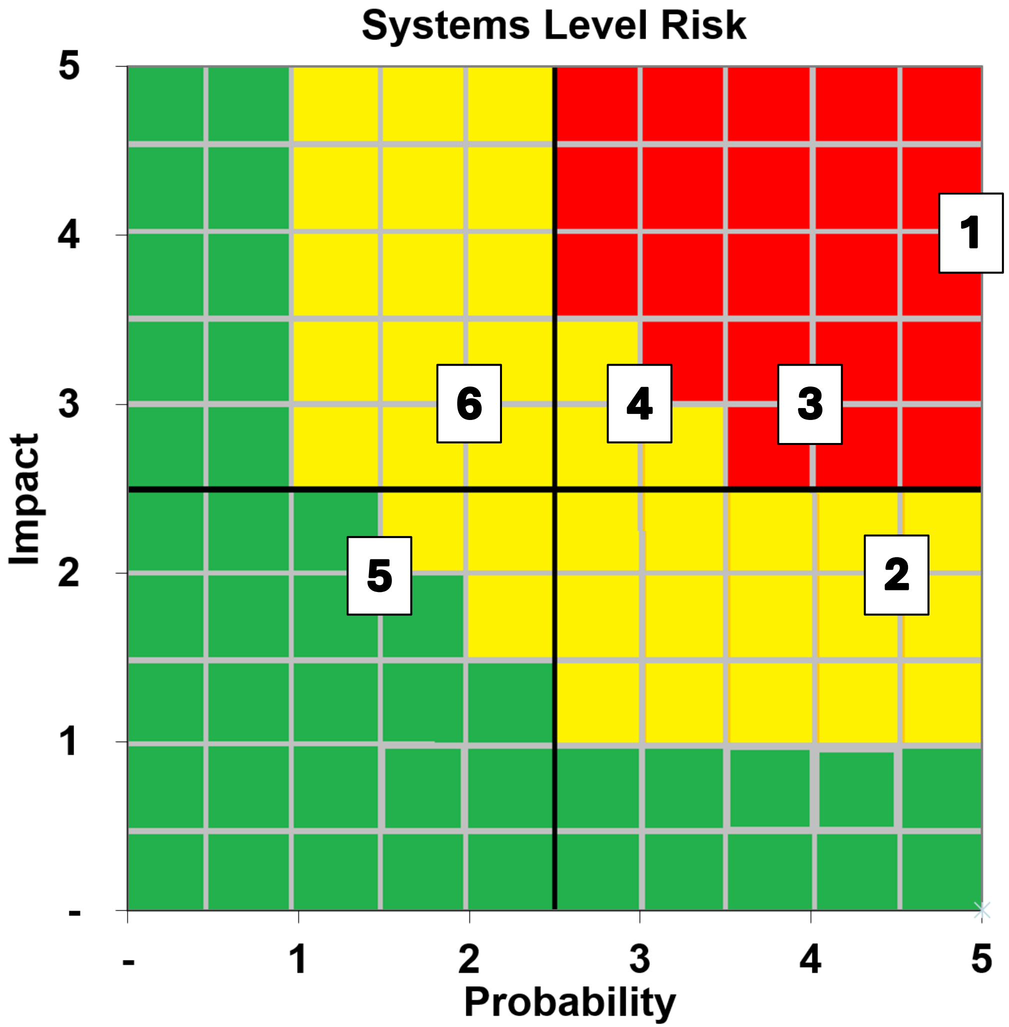
System Risks

Producibility

Budget

ODIN has Mitigated System Risks

1. Subsystem integration delays
2. Supply chain/cost increase
3. Sensor data degradation
4. Structural failure
5. Electromagnetic interference effects
6. Induced oscillation



ODIN is Assessed to be Producible

- Additive Manufacturing Producibility Reviews
  - 3-DOF Bearing Assembly (LMS)
  - Superstructure Corner Brackets (RPL)
  - Chassis (RPL)
- Assembly Reviews
  - Superstructure (RPL/Order)

Assembled Superstructure

*RPL=Rapid Prototyping Lab
*LMS = Light Manufacturing Shop
*AXFAB=Aerospace Experimentation and Fabrication

Structures Subsystem is Projected to be Within Budget and On Time

Part Name	Qty.	Status	Cost (USD)
Chassis	1		$0
Corner	6		$0
OAK-D S2 Sensor Bracket	1		$0
Linear Actuator Bracket	1		$0
Center Mount	1		$0
Center Dowel	1		$0
Slip Ring Mount	4		$0
Exterior Ring	1		$0
Suspension Mount to T-slot	3		$0
36" 8020 Aluminum Extrusion - 47065T102	9		$248.22
30"L Eye-to-Eye Lanyard - 30645T392	3		$45.33
Screw-In Hook - 29905T45	3		$29.40

Not Submitted

In Manufacture or Ordered

Complete or in hand

Structures Subsystem is Projected to be on Budget and On Time Continued

Part Name	Qty.	Status	Cost (USD)
Pulley - 3213T59	3		$12.65
T-Slotted Framing Fasteners - 4976N32	1		$3
8"L Eye-to-Eye Lanyard - 30645T392	1		$8.94
1/4"-20 Screw Hanger - 90172A545	1		$18.37
One-Piece Ball Thrust Bearing - 60715K15	1		$24.21
Slip Ring	3		$23
Tapered Heat Threaded Insert Kit M2/3/4/5	1		$13.90
20mm x 8mm Steel Dowel - 91595A558	1		$8.92
14mm x 8mm Steel Dowel - 91595A550	1		$8.03
5mm Collet	1		$6.99
5mm x 40mm rod	1		$6.99
5mm to 1/4 inch coupler	1		$8.49

Not Submitted

In Manufacture or Ordered

Complete or in hand

Controls Subsystem is Projected to be Within Budget and On Time

Part Name	Qty.	Status	Cost (USD)
250mm Linear Acutuator + Nema 14 Motor	3		$0
50mm Linear Acutuator + Nema 14 Motor	1		$85.00
Pololu Motor	4		$0
Super Precision Gyroscope	4		$0
Nema 23 Stepper Motor	4		$0
250mm Linear Acutuator Weight + Attachment	3		$0
50mm Linear Actuator Weight + Attachment	1		$0

Not Submitted
In Manufacture or Ordered
Complete or in hand

Power Subsystem is Projected to be Within Budget and On Time

Part Name	Qty.	Status	Cost (USD)
Power Switch	2		$5.26
Pololu Motor Driver Board	4		$51.80
TMC5160SilentStepStick Stepper Controller	8		$51.28
Battery Management System	1		$7.49
12V Step Down Converter	1		$0
5V Step Down Converter	1		$15.95
6A Light Up Circuit Breaker	1		$6.67
10A Light Up Circuit Breaker	1		$7.30
Heat Sink w/ Thermal Adhesive	15		$12.90
8-Channel Level Shifter	1		$7.95
Screw Terminal Pack (Blue)	1		$8.45
2.5A Circuit Breaker	2		$20.54

Not Submitted

In Manufacture or Ordered

Complete or in hand

Command Subsystem is Projected to be Within Budget and On Time

Part Name	Qty.	Status	Cost (USD)
Rasberry Pi 5/16GB	1		$120.00
Rasberry Pi Active Cooler	1		$10.95
VN-200S Rugged	1		$0
Slamstick	1		$0
Raspberry Pi Pico	1		$3.95

Not Submitted
In Manufacture or Ordered
Complete or in hand

Sensor Subsystem is Projected to be Within Budget and On Time

Part Name	Qty.	Status	Cost (USD)
OAK-D S2	1		$250.00

Not Submitted
In Manufacture or Ordered
Complete or in hand

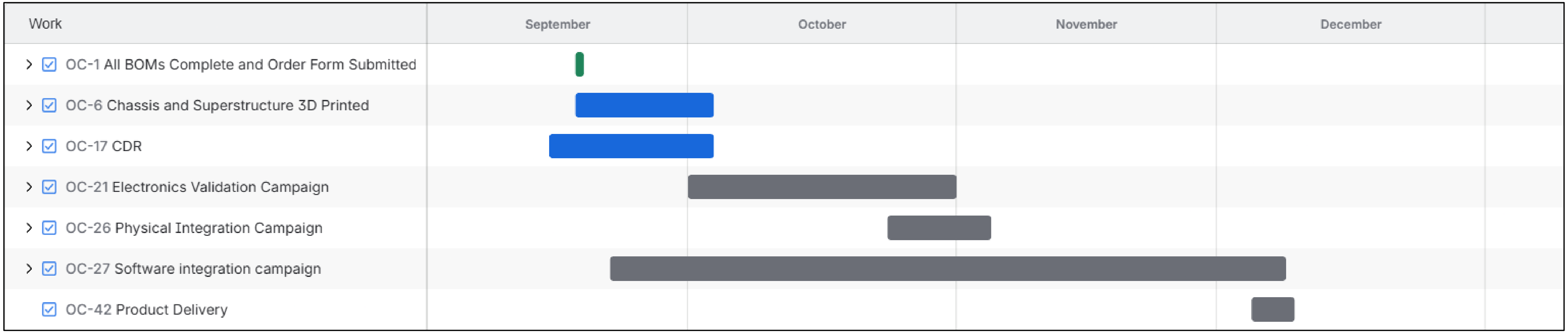
ODIN is Projected to be Within Budget and On Time

Subsystem	Status	Cost (USD)
Structures		$544.62
Controls		$85.00
Power		$195.59
Command		$134.90
Sensor		$250.00
	Total	**$1210.11**

Not Submitted
In Manufacture or Ordered
Complete or in hand

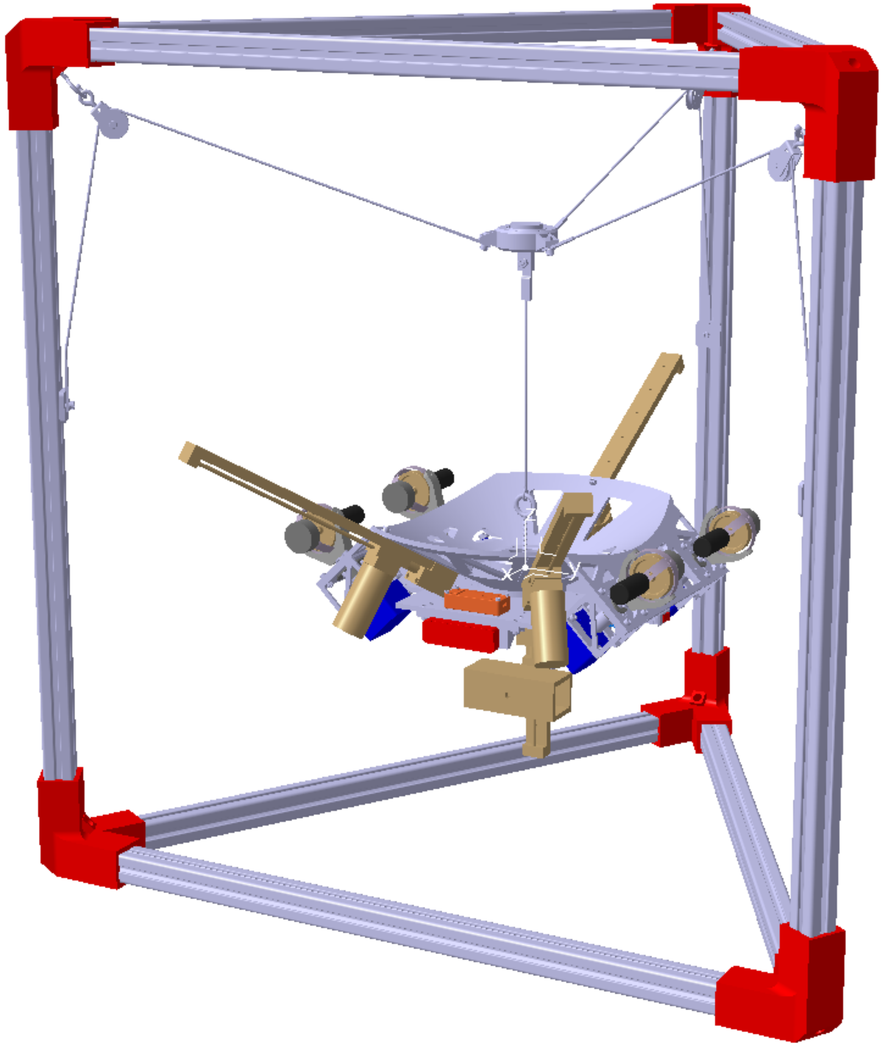
ODIN is Projected to be On Time

- Atlassian Jira used to document and track activities
- Major milestones to be incremental
- High schedule risk work to be done in parallel



ODIN is Projected to Meet Requirements

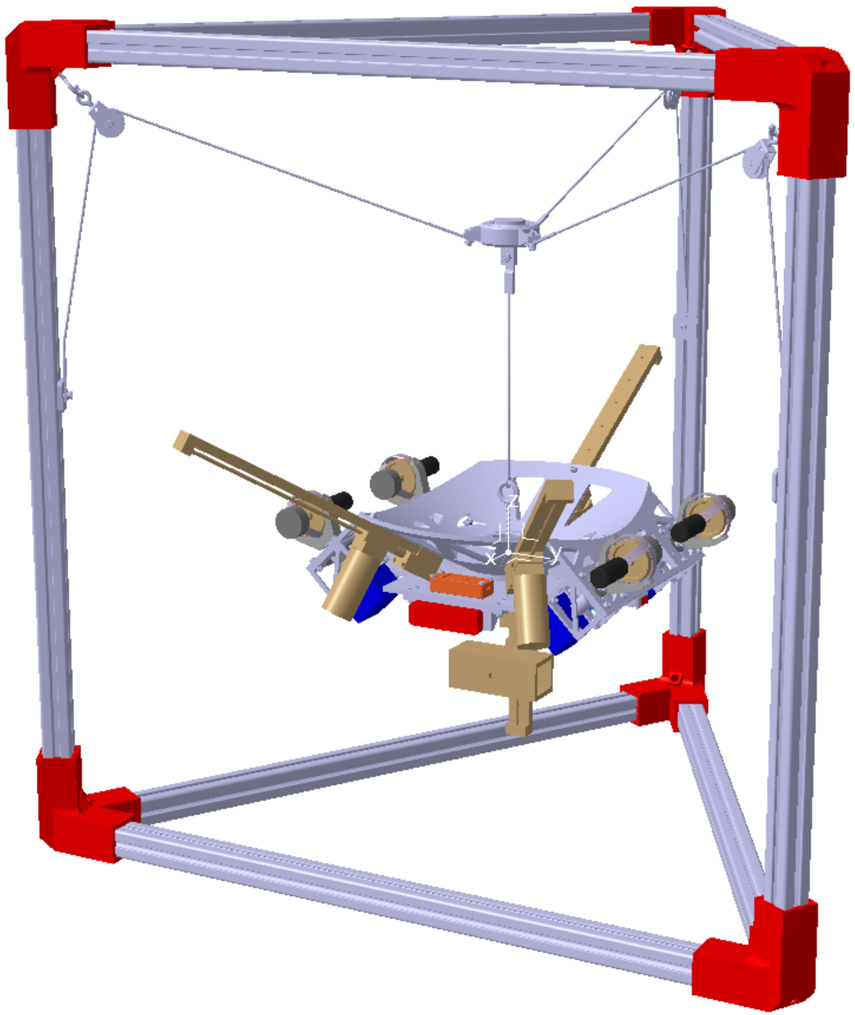
Req ID	Requirement
OBJ.2	The system should calibrate mass distribution properties.
OBJ.3	The system should be transportable by no more than two people.
OBJ.4	The system should record critical data onboard for diagnostics.
OBJ.6	The system should be capable of tracking nearby objects.
OBJ.7	The system should operate disconnected from external power.
OBJ.8	The system should determine target state information.

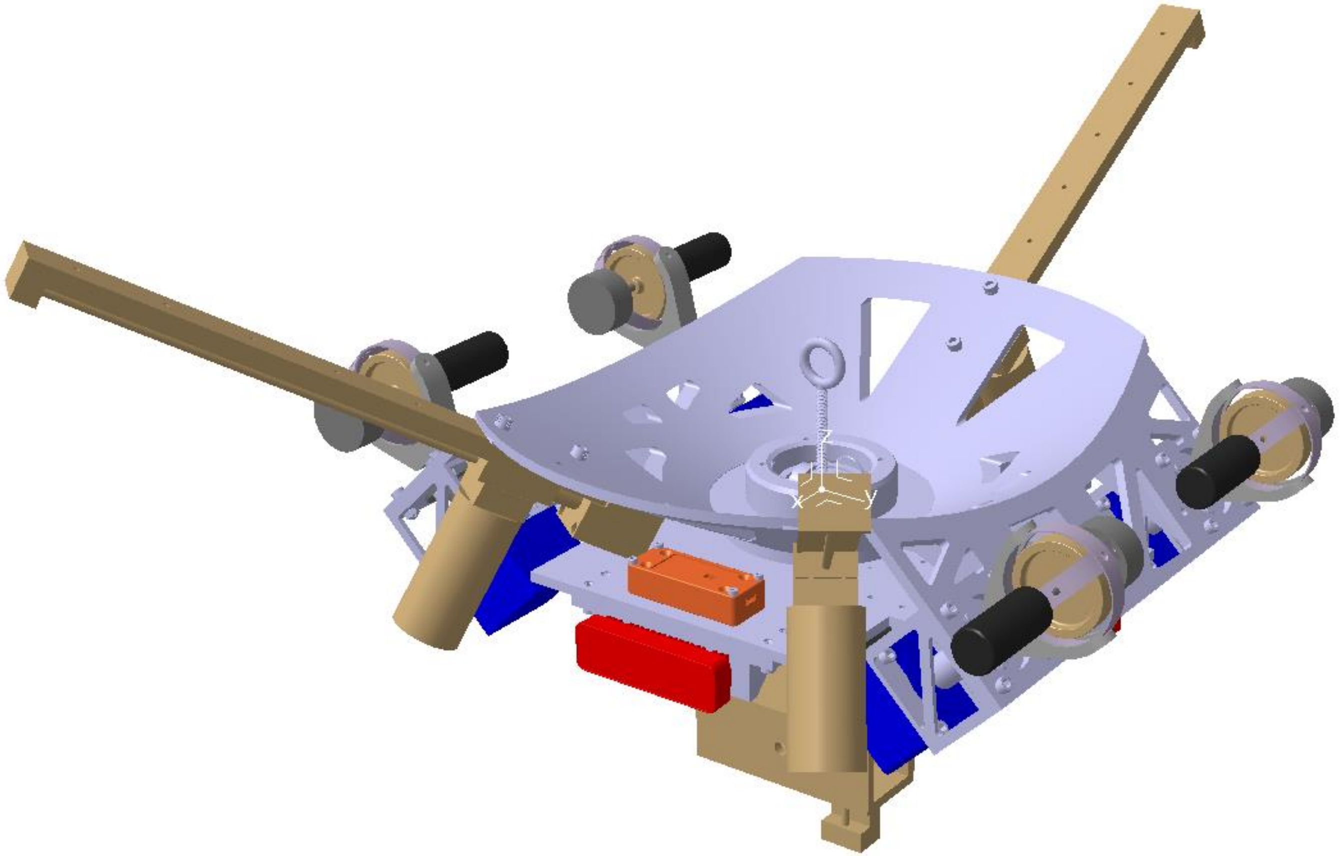


Acknowledgments

- Special thanks to Dr. Bradley Wall, Dr. Ron Madler, Dr. Kathy Wood, Dr. Rivera Zapata, Dr. Seth McNeill, Professor Joseph Chandler, and Professor Kathryn Wesson for technical insight and review of this program.

Questions





JIC

References

[1] NASA, "Space Debris and Human Spacecraft," 2021. Available: https://www.nasa.gov/mission_pages/station/news/orbital_debris.html.

[2] U.S. Space Force, "Space Surveillance Network Fact Sheet," 2020. Available: https://www.spaceforce.mil/factsheets/SSN.

[3] D. Kessler and B. Cour-Palais, "Collision Frequency of Artificial Satellites: The Creation of a Debris Belt," Journal of Geophysical Research, vol. 83, no. A6, pp. 2637-2646, 1978.

[4] EatYourBites, “List of gpus by Generative Ai Tops,” List of GPUs by Generative AI TOPS. Available: https://www.eatyourbytes.com/list-of-gpus-by-generative-ai-tops/.

[5] Gyroscope.com, Super Precision Gyroscope Manual, Version 9.0, Gyroscope.com, April 2018. Available: https://www.gyroscope.com/images/super/SuperManual9C-EN.pdf

[6] FUYU Technology Co., Ltd., FSK30J Series Micro Ball Screw Actuator Linear Motion Guide with Stepper Motor, Chengdu, China, 2025. Available: https://www.fuyumotion.com/miniature-aluminum-profile-small-and-light-linear-rail-guide-with-stepper-motor-product/

[7] Wall, B., Satellite Attitude Dynamics Control, unpublished manuscript, Embry-Riddle Aeronautical University, Prescott, AZ, 2025.

[8] Pololu, “Pololu - 5V, 3.4a step-down voltage regulator D30V30F5,” Pololu Robotics & Electronics. Available: https://www.pololu.com/product/4892.

[9] U.S. Department of Defense, Environmental Engineering Considerations and Laboratory Tests, MIL-STD-810H, Washington, D.C., 2019.

[10] ASTM International, Standard Test Method for Tensile Properties of Plastics, ASTM D638-14, West Conshohocken, PA, 2014.

[11] ASTM International, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes, ASTM B221-14, West Conshohocken, PA, 2014.

[12] Society of Automotive Engineers, Test Requirements for Linear Mechanical Actuators, SAE ARP5812, Warrendale, PA, 2011.

[13] International Organization for Standardization, Safety of Machinery – Safety-Related Parts of Control Systems – Part 1: General Principles for Design, ISO 13849-1:2015, Geneva, Switzerland, 2015.

[14] CubeSat Program, CubeSat Design Specification, Rev. 14.1, California Polytechnic State University, San Luis Obispo, CA, 2020. [Includes AFSPCMAN91-710V3 citation]

[15] Institute of Electrical and Electronics Engineers, Standard for Advanced Audio and Video Coding, IEEE 1857-2016, New York, NY, 2016.

[16] International Organization for Standardization, 3-D Scanning Methodologies for Body Measurements, ISO 20685-1:2018, Geneva, Switzerland, 2018.

[17] Institute of Electrical and Electronics Engineers, Sensor Performance Parameter Definitions, IEEE 2700-2017, New York, NY, 2017.

[18] Institute of Electrical and Electronics Engineers, Time-Sensitive Networking Standard, IEEE 802.1Q-2020, New York, NY, 2020.

[19] IPC, Generic Standard on Printed Board Design, IPC-2221B, Northbrook, IL, 2012.

[20] European Cooperation for Space Standardization, Space Product Assurance – Electrical, Electronic and Electromechanical (EEE) Components, ECSS-Q-ST-60C, Noordwijk, The Netherlands, 2008.

[21] NASA, Crimping, Interconnecting Cables, Harnesses, and Wiring, NASA-STD-8739.4A, Washington, D.C., 2016.

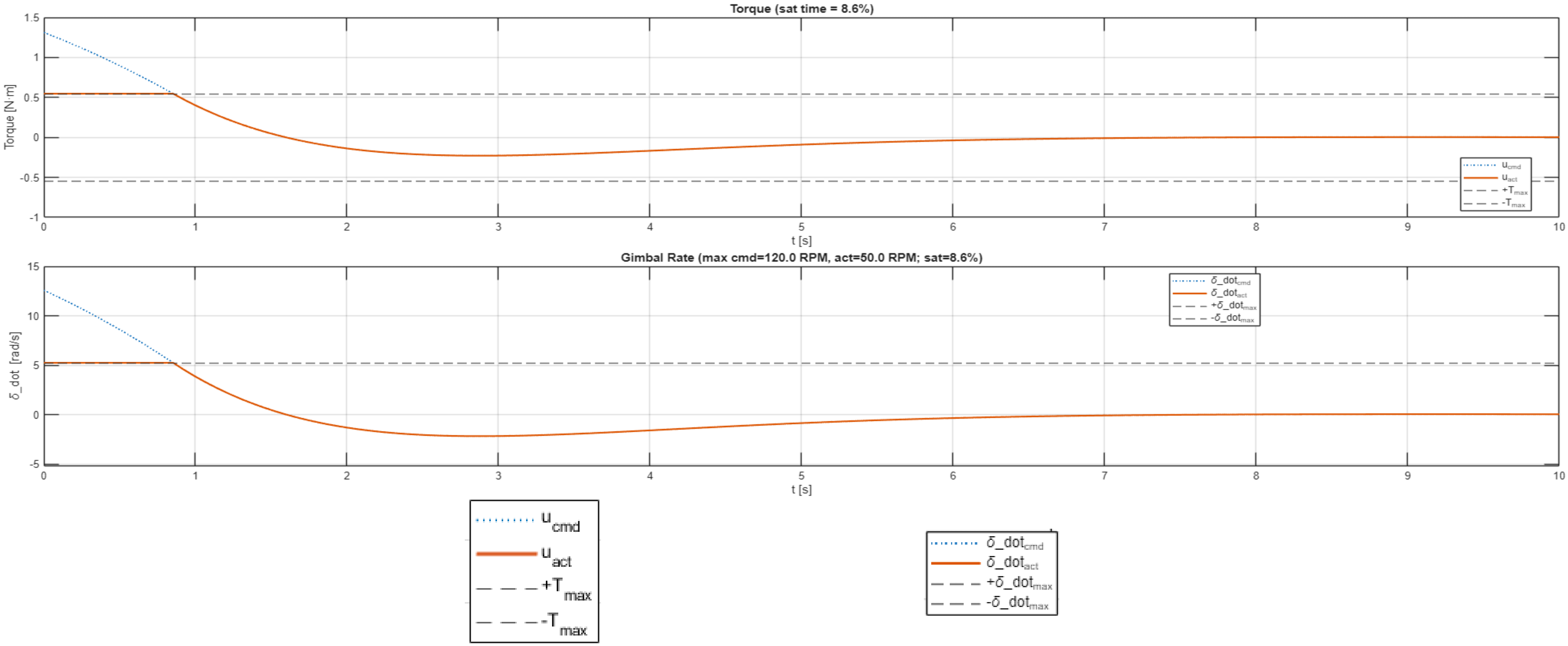
[22] International Electrotechnical Commission, Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems – General Requirements, IEC 61508-1:2010, Geneva, Switzerland, 2010.

[23] American Institute of Aeronautics and Astronautics, Guide to Safety of Lithium-Ion Batteries in Space Applications, AIAA G-136-2022, Reston, VA, 2022.

[24] European Cooperation for Space Standardization, Electrical and Electronic Design – Space Engineering, ECSS-E-ST-20-20C, Noordwijk, The Netherlands, 2008.

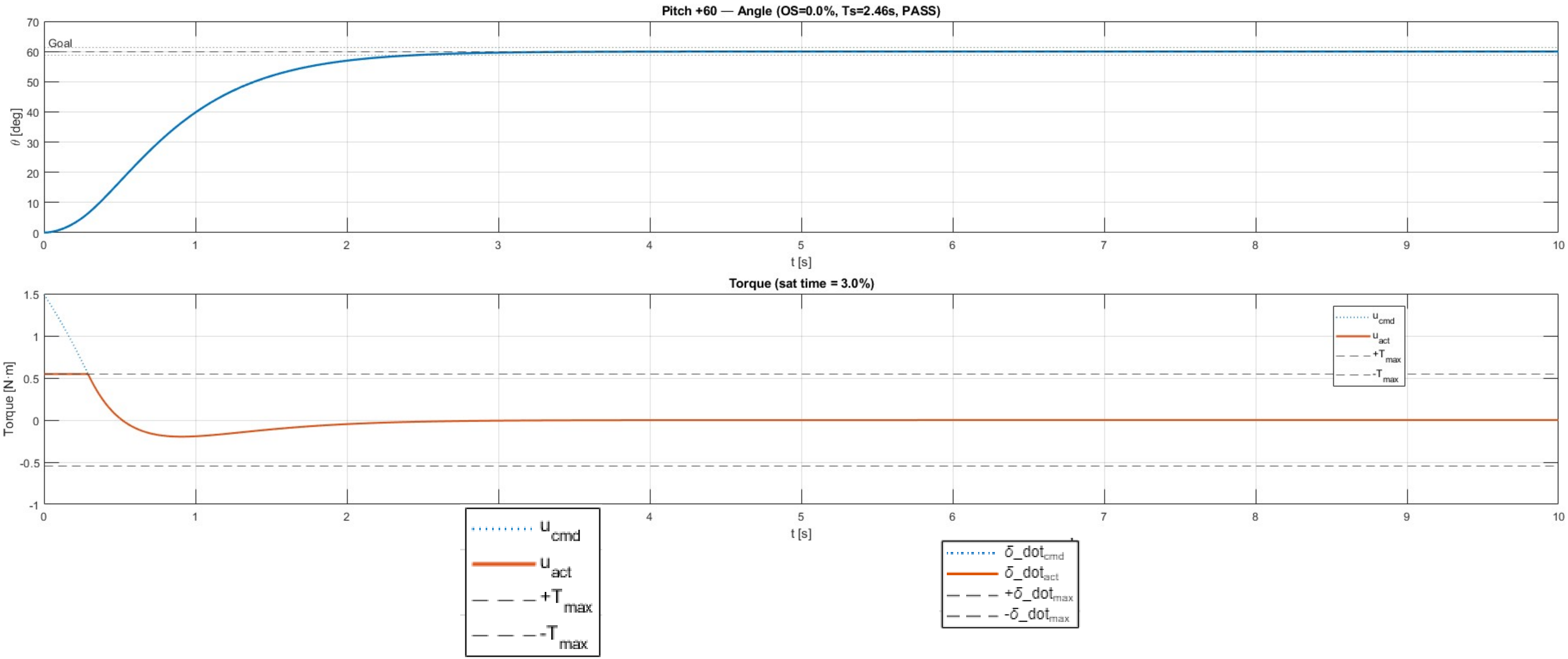
[25] Society of Automotive Engineers, Spacecraft Electrical Power Quality Requirements, SAE AS5698A, Warrendale, PA, 2019.

Yaw Results

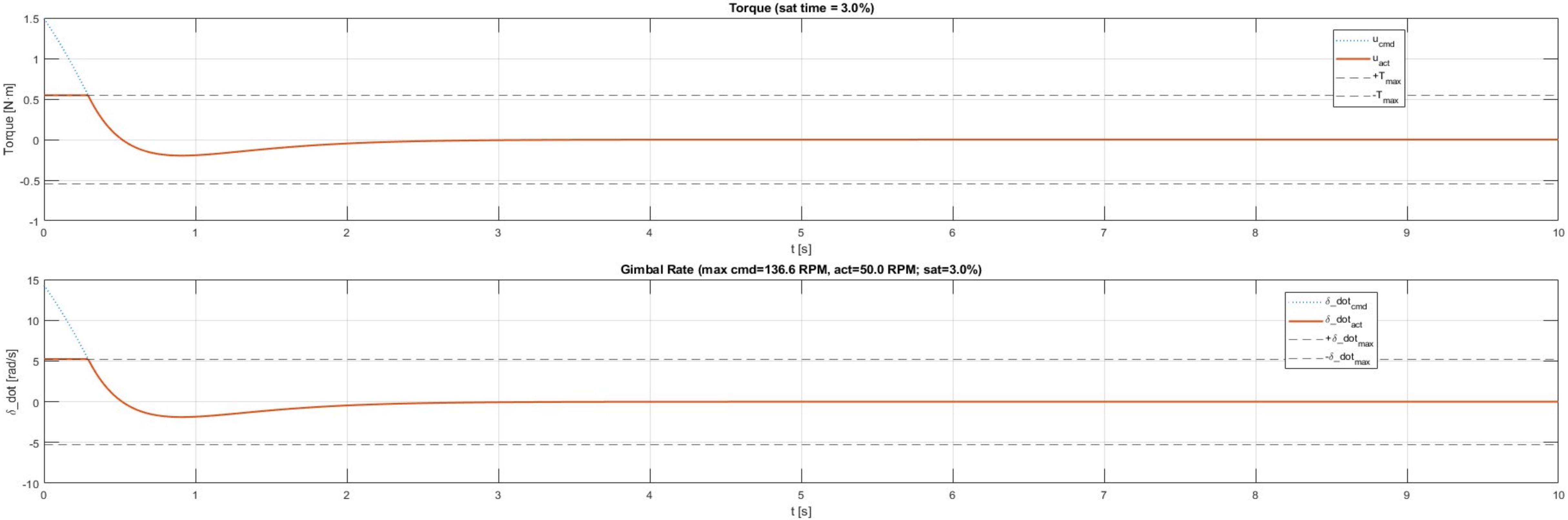


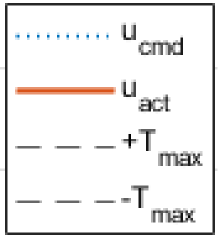
Torque (sat time = 8.6%)
Torque [N·m]
t [s]
u_{cmd}
u_{act}
$+T_{max}$
$-T_{max}$
Gimbal Rate (max cmd=120.0 RPM, act=50.0 RPM; sat=8.6%)
δ_dot [rad/s]
t [s]
δ_dot$_{cmd}$
δ_dot$_{act}$
+δ_dot$_{max}$
-δ_dot$_{max}$

Pitch Results

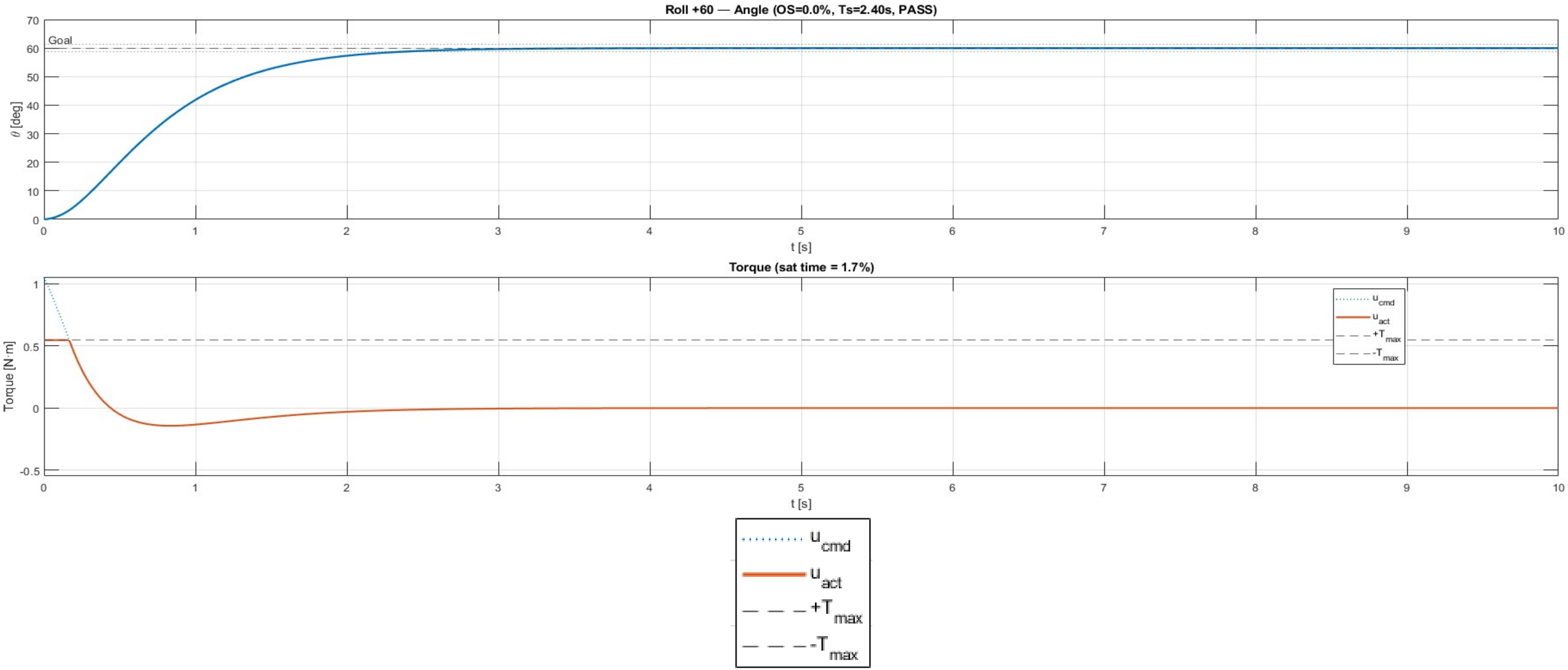


Pitch Results

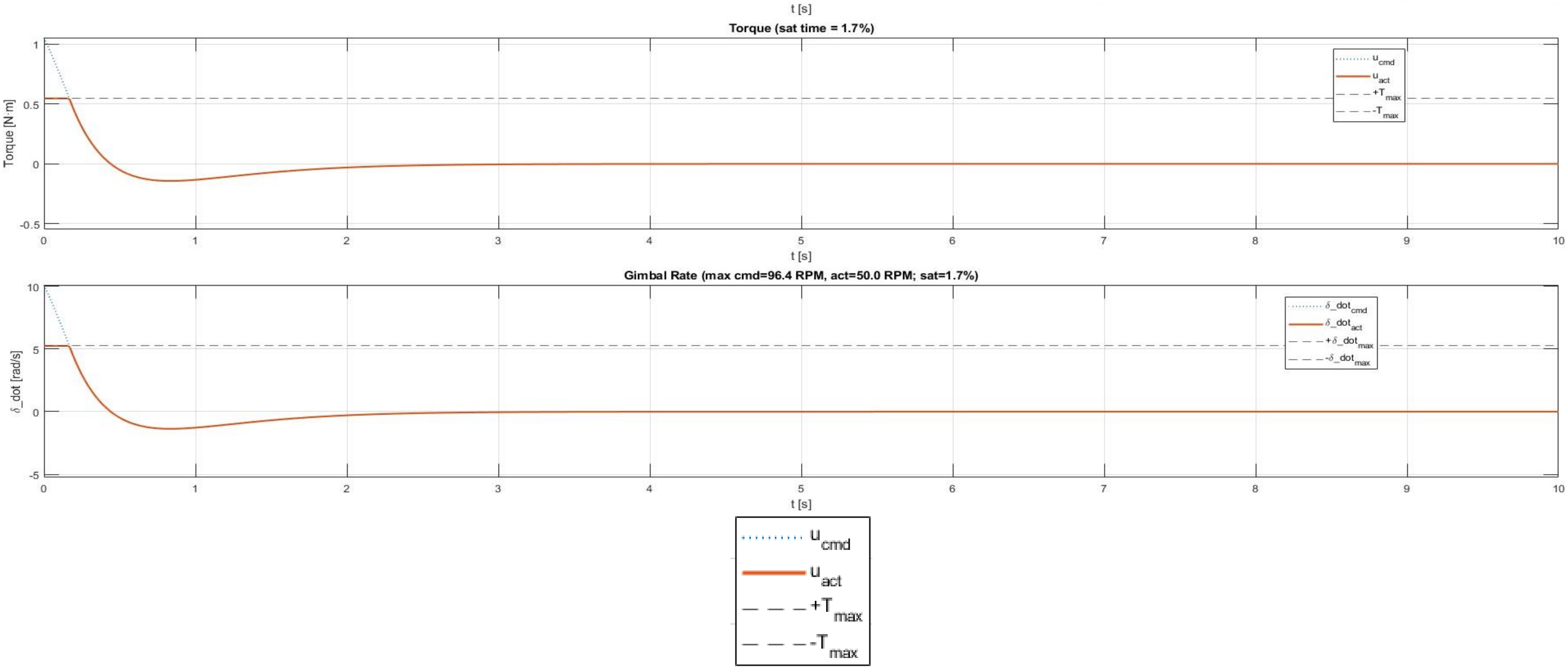




Roll Results



Roll Results



ODIN Mass Cost and Power Table

SUBSYSTEM	ASSEMBLY	SIZE (mm)	MASS (kg)	POWER (W)	COST ($USD)
Structures	Superstructure	900x900x900	5	0	$300
	Frame	300x300x350	0.5	0	$15.00
Controls	Linear Actuator	387x27x27	3.3	34.2	$90.00
	CMG Assembly	62.5x133x184	3.3	10	$-
	Control Board	25.4x25.4x7.62	0.00022	4	$73.08
Optics	Oak D - S2	54.5x33x110	0.09	5	$249.00
Computation	Microcontroller	101x53	0.075	15	$106.59
	IMUs	25x15x5, 72x25x18	0.007	15	$-
Power	Storage	166x175x43	0.635	N/A	$-
		Totals	**12.91**	**83.20**	**$833.67**
		Margin	14.20	91.52	$917.04

Testbed Will be Hanging From a Superstructure

DECISION CRITERIA		Mounting Options								
		TIPSY			HANGER (SINGLE STRING)			HANGER (MULTI STRING)		
Must Haves		**Info**		**Y/N**	**Info**		**Y/N**	**Info**		**Y/N**
Stable Rotation With No Translation		No Translation		Yes	Potential to Swing		No	Triangulated with No Swing		Yes
Wants	**Weight**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**
Freedom of Motion (degrees)	1	Degrees of Pitch	35	35	Degrees of Pitch	75	75	Degrees of Pitch	75	75
Cost ($)	1	Use Existing Frame Hardware	0	0	Simple cable	21.99	21.99	Triangulated cable w/ thrust bearing	25.91	25.91
Relative Merit				**35**			**75**			**75**
Normalized Merit				35			75			75
Normalized Cost				0			21.99			25.91
Merit/Cost				**35**			**53.01**			**49.09**
SELECTED COMPONENT								✓		

Structures Subsystem Trade Study: Mounting Architecture

Chassis Will be Additively Manufactured out of PETG

DECISION CRITERIA										OI
		PLA			ABS			PAHT-CF (Carbon Fiber Nylon)		
Must Haves		Info		Y/N	Info		Y/N	Info		Y/N
Young's Modulus >= 1 GPa		2.58 GPa		Yes	2.2 GPa		Yes	3.86 GPa		Yes
Wants	Weight	Info	Value	Score	Info	Value	Score	Info	Value	Score
Density (g/cc)	7	Low Density Plastic	1.24	8.68	Low Density Plastic	1.05	7.35	Low Density Plastic	1.06	7.42
Manufacturability (1-10)	9	3D Printing	10	90	3D Printing	9.5	85.5	3D Printing	8.5	76.5
Tensile Strength (X-Y) (MPa)	1	Tensile Strength in-layer	35	35	Tensile Strength in-layer	33	33	Tensile Strength in-layer	92	92
Tensile Strength (Z) (MPa)	1	Tensile Strength between layers	31	31	Tensile Strength between layers	28	28	Tensile Strength between layers	47	47
Young's Modulus (GPa)	10	Extensional Stiffness	2.58	25.8	Extensional Stiffness	2.2	22	Extensional Stiffness	3.86	38.6
Cost ($/kg)	4	Cost per kg	19.99	79.96	Cost per kg	19.99	79.96	Cost per kg	95.99	383.96
Relative Merit				**190.48**			**175.85**			**261.52**
Normalized Merit				28.47			30.89			26.66
Normalized Cost				4.00			4.00			19.20
Merit/Cost				**24.48**			**26.89**			**7.46**
SELECTED COMPONENT										

J - Equation for Merit Normalization
J = 7*(1/Density) + 9*(1/10)*Manufacturability + 1*(1/30)*Z_Strength + 1*(1/30)*XY_Strength + 10*(3/Young)

J - Equation for Cost Normalization
J = 4*(1/20)*Cost

Structures Subsystem Trade Study: Material Selection

Chassis Will be Additively Manufactured out of PETG

PETG			TPU			6061-T6 Aluminum			304 Stainless Steel		
Info		Y/N	Info		Y/N	Info		Y/N	Info		Y/N
1.81 GPa		Yes	0.0098 GPa		No	68.9 GPa		Yes	193 GPa		Yes
Info	Value	Score	Info	Value	Score	Info	Value	Score	Info	Value	Score
Low Density Plastic	1.28	8.96	Thermo-Plastic	1.22	8.54	Higher Density Metal	2.7	18.9	Highest Density	8	56
3D Printing	10	90	3D Printing (slow)	9	81	Machining	4	36	Machining	5	45
Tensile Strength in-layer	34	34	Tensile Strength in-layer	27.3	27.3	Tensile Strength	310	310	Tensile Strength	505	505
Tensile Strength between layers	23	23	Tensile Strength between layers	22.3	22.3	N/A	0	0	N/A	0	0
Extensional Stiffness	1.81	18.1	Extensional Stiffness	0.0098	0.098	Extensional Stiffness	68.9	689	Extensional Stiffness	193	1930
Cost per kg	19.99	79.96	Cost per kg	41.99	167.96	Cost per kg	2.5	10	Cost per kg	2.8	11.2
		174.06			**139.24**			**1053.90**			**2536.00**
		32.94			3076.72			16.96			22.36
		4.00			8.40			0.50			0.56
		28.95			**3068.32**			**16.46**			**21.80**
✓			Potential to Integrate into Structure for Vibrational Damping								

Structures Subsystem Trade Study: Material Selection

Superstructure Will be Constructed From Aluminum T-Channel

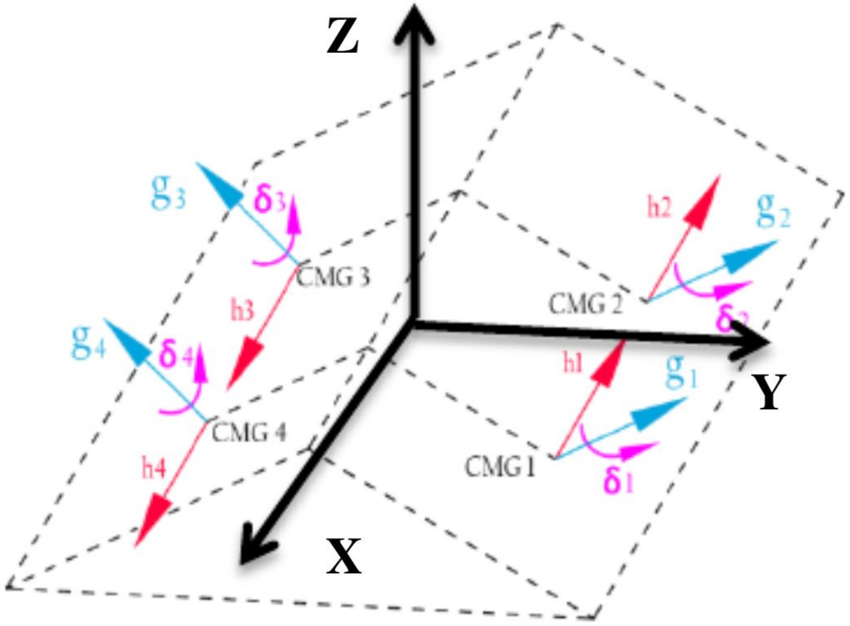
DECISION CRITERIA		ODIN Frame Material Composition OPTIONS											
		Aluminum (Machined)			Aluminum T-Channel			Steel (Machined)			Steel (Waterjet Cut)		
Wants	Weight	Info	Value	Score	Info	Value	Score	Info	Value	Score	Info	Value	Score
Ease of Construction (1-10)	10	Clevis and Hole Construction	10	100	T-Nut and T-Slot	9.5	95	Clevis and Hole Construction	10	100	Clevis and Hole Construction	8	80
Manufacturability (1-10)	7	Machining Required	7	49	3D Printing for Joints	10	70	Machining Required	7	49	Mild Machining/Rapid Waterjet	9	63
Transportability (1-10)	9	Smaller Parts to Track	9	81	Easy Pre-Assembled Sections	10	90	Smaller Parts to Track	8	72	Waterjet Parts May Be Bulky	6	54
Cost ($/stock)	8	Price per 2 Feet of 1/2" Round 6061	6.02	48.16	Cost per 1 Foot Section of 1"x1" 80-2(	5.5	44	Price per 2 Feet of 1/2" Round A-36	4.4	35.2	Price per 2 Feet of 1/2" Round A-36	4.4	35.2
Relative Merit				230			255			221			197
Normalized Merit				26			29.5			25			23
Normalized Cost				8.03			7.33			5.87			5.87
Merit/Cost				17.97			22.17			19.13			17.13
SELECTED COMPONENT					✓								

J - Equation for Merit Normalization
J = 10*(1/10)*EaseOfConstruction + 7*(1/7)*Manufacturability + 9*(1/9)*Transportability

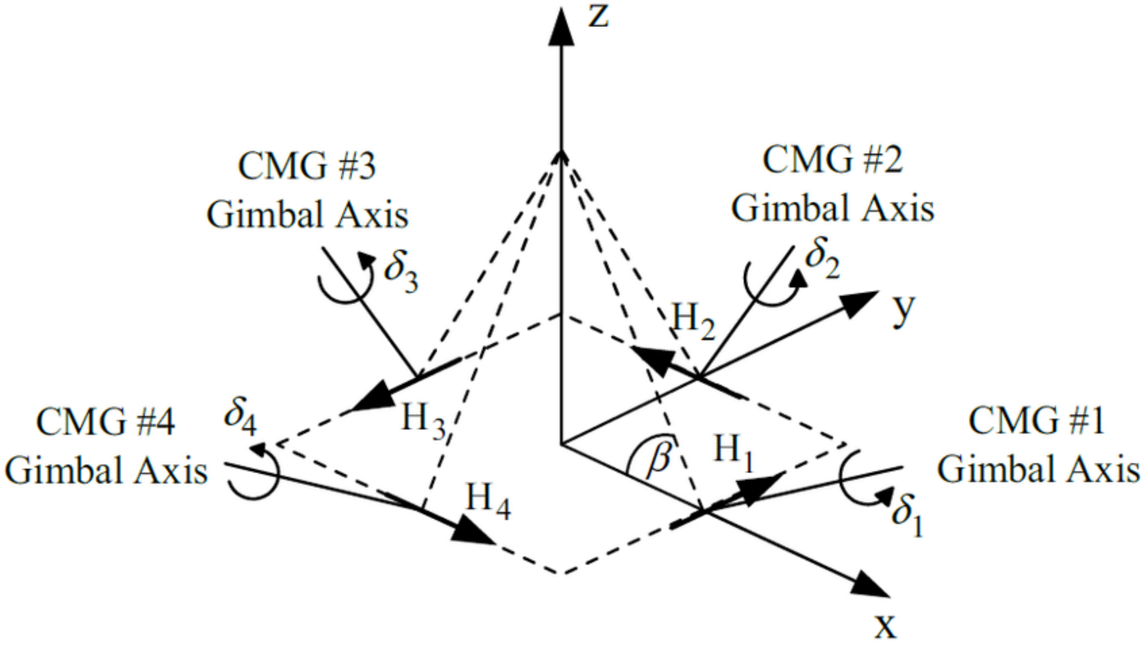
J - Equation for Cost Normalization
J = 8*(1/6)*Cost

Structures Subsystem Trade Study: Superstructure Selection

Comparison of CMG Cluster Configurations



Roof CMG Cluster



Pyramid CMG Cluster

Image Credit: Mony, Abhilash & Hablani, Hari & Sharma, Gireesh. (2016). Control Moment Gyro (CMG) Sizing and Cluster Configuration Selection for Agile Spacecraft.

Roof Array is the Superior CMG Cluster Configuration

DECISION CRITERIA		CMG Array Options					
		Roof Array			Pyramid Array		
Must Haves		Info		Y/N	Info		Y/N
Singularity avoidance		Easier		Yes	Harder		Yes
Control Law Simplicity		Easier		Yes	Harder		Yes
Wants	Weight	Info	Value	Score	Info	Value	Score
Pointing Accuracy (μrad)	8	22.48	7.02	56.16	22.33	7.02	56.16
Jitter (mrad)	5	5.52	4.48	22.4	5.22	4.78	23.9
Pointing Stability (mrad)	3	1.335	5.55	16.65	1.226	5.91	17.73
Ease of Integration	4	Less singularities	8	32	Harder singularity avoidance	4	16
Torque Capacity (Nm)	4	1	1	4	1.5	1.5	6
Industry representation	3	Less common	5	15	More widely used	8	24
Relative Merit				146.21			143.79
Normalized Merit				4.87			4.79
Merit				4.87			4.79
SELECTED COMPONENT		✓					

J - Equation for Merit Normalization
J = 8*Pointing Accuracy/3.05 + (10-Jitter)*5 + Pointing Stability*3*3 + Integration*4 +Torque*4 +Industry representation*3

Center of Mass Will Be Controlled Using RATTMMOTOR CBX1605 100mm Linear Actuator

DECISION CRITERIA		CMG Array Options								
		RATTMMOTOR EBX1605 400mm			Mini Linear Lead Screw			RATTMMOTOR CBX1605 100mm		
Must Haves		Info		Y/N	Info		Y/N	Info		Y/N
Precision Control				Yes			Yes			Yes
Software Compatability				Yes			Yes			Yes
Wants	Weight	Info	Value	Score	Info	Value	Score	Info	Value	Score
Positional Accuracy (mm)	8	0.01	9	72	0.025	7.5	60	0.03	7	56
Energy Cinsumption (W)	4	7.2	2.8	11.2	5.4	4.6	18.4	7.2	2.8	11.2
Total Weight(g)	4	4100	7.820553085	31.28221	55	3.432256	13.72902	1678	12.84593	51.38373
Force Output (N)	3	98	9.8	29.4	10	1	3	98	9.8	29.4
Stroke Length (mm)	6	400	5.333333333	32	34	0.453333	2.72	100	1.333333	8
Price ($)	7	107.58	1.859081614	13.01357	26.83	7.454342	52.1804	69.96	2.858776	20.01144
Relative Merit				143.8822123			95.12902411			147.9837328
Normalized Merit				4.80			3.17			4.93
Merit				4.80			3.17			4.93
SELECTED COMPONENT									✓	

J - Equation for Merit Normalization
J = 7*(10-Accuracy*100) + 4(1000*e^(-abs(2000-weight)^2*30) + 3*(Force/10) + 6*(Stroke Length/75) + 7*(200/Price)

ODIN Will Use a Moore-Penrose Pseudoinverse Control Law

DECISION CRITERIA		CMG Array Options								
		Moore-Penrose			Singularity Robust Inverse			Local gradient		
Must Haves		Info		Y/N	Info		Y/N	Info		Y/N
Singularity avoidance				Yes			Yes			Yes
Computational Feasibility				Yes			Yes			Yes
Wants	Weight	Info	Value	Score	Info	Value	Score	Info	Value	Score
Singularity Avoidance Effectivness	8		5	40		8	64		10	80
Computational Efficiency	5		9	45		8	40		7	35
Accuracy Tracking	6	Accuracy degredation towards singularities	7	42		8	48		10	60
Ease of Integration	9		9	81		6	54		3	27
Relative Merit				**208**			**206**			**202**
Normalized Merit				6.93			6.87			6.73
Merit				**6.93**			**6.87**			**6.73**
SELECTED COMPONENT		✓								

J - Equation for Merit Normalization

J = Avoidance*8 + Computational Efficiency*5 + Accuracy*6 + Integration*9

ODIN will use an OAK-D S2

DECISION CRITERIA		ODIN Optical Sensor Options											
		Nicla Vision			OpenMV Cam RT1062			OAK-D S2			Intel RealSense D455		
Must Haves		Info		Y/N	Info		Y/N	Info		Y/N	Info		Y/N
Can Track Position				Yes			Yes			Yes			Yes
Can Track Velocity				Yes			Yes			Yes			Yes
Wants	**Weight**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**
Frame rate (Hz)	10		8	80		30	300		120	1200		90	900
Arduino/PI Integration	9		1	9		1	9		1	9		1	9
Resolution (MP)	1		2	2		5	5		1	1		1	1
Power Required (W)	2		0.525	1.05		0.65	1.3		5	10		4	8
Computational Cost	10	extra needed	0.5	5	extra needed	0.5	5	onboard computation	1	10	extra needed	0.5	5
FOV (degrees)	6	horizontal	80	480	horizontal	70.8	424.8	horizontal	80	480	horizontal	87	522
Cost ($)	2		120	240		120	240		249	498		239	478
Relative Merit				**817.05**			**985.10**			**2208.00**			**1923.00**
Normalized Merit				109.02			114.16			119.68			112.17
SELECTED COMPONENT								✓					

J - Equation for Merit Normalization
J = 10*(Frame rate/100) + 9*(10)*integration + Resolution + 2/power + 10*computation cos + 6*FOV/180 + 2 *(450/cost)

Optics Subsystem Trade Study: Component Selection

OAK-D S2 Key Features

- 3d position/velocity detection
- 120 Hz (greyscale)
- 1 megapixel
- Onboard computing
- 80 degrees horizontal FOV
- $249 cost
- 5 W consumption

ODIN will use a combination of existing IMUs

DECISION CRITERIA		IMU OPTIONS (Variable Configurations)								
		Existing Corals/TIPSY Hardware (MPU9250/6500)			**ENDAQ Slamstick**			**Adafruit 9-DOF BNO055**		
Must Haves		**Info**		**Y/N**	**Info**		**Y/N**	**Info**		**Y/N**
Com Interface (pinout/usb)		I2C		Yes	USBC		Yes	I2C		Yes
Data output rate (>=100Hz)		100Hz		Yes	3200-20,000		Yes	100		Yes
Wants	**Weight**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**
Data output format (quaternion/Euler)	2	Raw Data	1	2	Raw Data	1	2	Quaternion and Euler	3	6
Documentation	2	HiLetgo	3	6	EnDAQ Commerical Product	10	20	Adafruit	10	20
Size (cm)	3	2.5*1.5	5	15	7.2*2.5*1.8	1	3	2.6*2	5	15
Cost (US Dollars)	1	Already Owned	100	100	Already Owned	100	100	Adafruit Store	29.95	29.95
Relative Merit				**123**			**125**			**70.95**
Normalized Merit				0.93			0.9967			1.867
Normalized Cost				5			5			1.498
Merit/Cost				**5.93**			**6.00**			**3.364**
SELECTED COMPONENT		✓			✓					

J - Equation for Merit Normalization
J = 2*(1/10)*Data_Output _Format + 2*(1/30)*Documentation + 3*(1/30)*Size
J - Equation for Cost Normalization
J = 1*(1/20)*Cost

ODIN will use an Arduino GIGA R1

DECISION CRITERIA		MICROCONTROLLER OPTIONS								
		Arduino Due (TIPSY)			Arduino GIGA R1 WiFi			PIC32MZ-W1		
Must Haves		**Info**		**Y/N**	**Info**		**Y/N**	**Info**		**Y/N**
I/O Pin Count > 30		54		Yes	76		Yes	20* (modular)		Yes
Wireless Module		N/A		No	Wifi		Yes	Wifi		Yes
RAM >1 MB		512 kB		No	3 MB		Yes	1 MB		Yes
Wants	**Weight**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**	**Info**	**Value**	**Score**
Documentation	3	Arduino	1	3	Arduino	1	3	Microchip	0.75	2.25
Clock Speed	5	84 MHz	84	420	480 MHz	480	2400	8 MHz	8	40
Interface (HW)	5	UART,SPI	2	10	UART,I2C,SPI	3	15	SPI,ICSP	2	10
Interface (Program)	4	micro-USB	2	8	USB-C	3	12	HW	1	4
Cost	2	$48.40	48.4	96.8	$106.58	106.58	213.16	$84.00	84	168
Relative Merit				**441.00**			**2430.00**			**56.25**
Normalized Merit				23.62			33.38			17.85
Normalized Cost				9.68			21.32			16.80
Merit/Cost				**2.44**			**1.57**			**1.06**
SELECTED COMPONENT					✓					

J - Equation for Merit Normalization

$$J = Documentation + \log(Clock\ Speed) + Interface + Interface$$

J - Equation for Cost Normalization

$$J = Cost/10$$

ODIN will use a Molicel P28A Rechargeable Battery

DECISION CRITERIA		POWER SUPPLY OPTIONS											
		Molicel P28A			Molicel P42A			Samsung 35E			C40 (LiFePo)		
Must Haves		Info		Y/N	Info		Y/N	Info		Y/N	Info		Y/N
Rechargable		Yes		Yes	Yes		Yes	Yes		Yes	Yes		Yes
Safety		Safe		Yes	Safe		Yes	Safe		Yes	Safer than Li-ion		Yes
Wants	Weight	Info	Value	Score	Info	Value	Score	Info	Value	Score	Info	Value	Score
Discharge Rate (A)	5		35	175		45	225		8	40		50	250
Nominal Voltage (V)	2	2.5 - 4.2 V	3.6	7.2	2.5 - 4.2 V	3.6	7.2	2.5 - 4.2 V	3.6	7.2	2.5 - 3.7 V	3.2	6.4
Power Capability (W)	7		126	882		162	1134		28.8	201.6		160	1120
Capacity (Ahr)	3		2.8	8.4		4.2	12.6		3.5	10.5		20	60
Energy Density (Whr/kg)	4	Capacity (Whr) per kg	223	892	Capacity (Whr) per kg	226	904	Capacity (Whr) per kg	263	1052	Capacity (Whr) per kg	175	700
Cost ($)	5		0	0		7.99	39.95		8.99	44.95		9	45
Relative Merit				1964.60			2282.80			1311.30			2136.40
Normalized Merit				14.76			17.57			8.57			19.35
Normalized Cost				0.00			4.44			4.99			5.00
Merit/Cost				14.76			13.13			3.58			14.35
SELECTED COMPONENT		✓											

J - Equation for Merit Normalization
J = 5*(1/50)*Discharge Rate + 2*(1/3.6)*Nominal Voltage + 7*(1/162)*Power Capability +3*(1/20)*Capacity

J - Equation for Cost Normalization
J = 5*(1/9)*Cost

ODIN Current Draws at Max Electrical Loading

Subsystem	Component	Nominal Voltage (V)	Unit Max Current Draw (A)
Control	NEMA-14 Stepper Motor (x4)	12	1.4 (2-phase)
Control	NEMA-23 Stepper Motor (x4)	24	1.24 (2-phase)
Control	Pololu Encoded Motor (x4)	12	2.4 (startup) 0.3 (no-load)
Command	Raspberry Pi 5(x1)	5	2 (load) 0.5 (idle)
Sensor	OAK-D S2 Sensor (x1)	5-5.5	2

Molicel P28A Power Supply

$$V_{total} = NV_{cell} = 4\ cells\ * 3.6V = 14.4\ V$$

$$I_{\max continuous} = I_{PI\ 5} + I_{servos} + I_{OAK} + I_{Gimbal}$$

$$= 2A + (4 * 0.3\text{A}) + 2\text{A} + (4 * 1.244\text{A}) = 10.18\text{A}$$

$$Servos\ have\ tolerance\ of\ \pm 50\%\ at\ no\ load$$

$$Tolerances\ of\ \pm 5\%\ for\ stepper\ motors\ (from\ TMC5160\ Website)$$

$$tolerance = \pm\left[\left(0.5 * (4 * 0.3)\right) + \left(0.05 * (4 * 1.414A)\right)\right] = \pm 0.88\ A$$

$$tolerance\ \% = \frac{tolerance}{I_{\max continuous}} = \frac{\pm 0.88\ A}{10.18\ A} = \pm 0.867 = \pm 8.67\%$$

ODIN Power Capacity

$$Servo\ Motor\ Startup\ Draw\ (I_{start}) = 2.6\text{A}$$
$$Max\ Current\ Draw\ (I_{max}) = 10.18\text{A}$$
$$No\ load\ Current\ Draw\ (I_{idle}) = 1.7\text{A}$$
$$Center\ of\ Mass\ Calibration\ Current\ Draw\ (I_{COM}) = 6.16\text{A}$$
$$Servo\ Motor\ Startup\ Time\ (t_{start}) = 2\ sec$$
$$Center\ of\ Mass\ Calibration\ Time\ (t_{COM}) = 3\ sec$$
$$Maximum\ Load\ Estimated\ Time\ (t_{max}) = 10\ min$$

Electric Charge Available

$$C = 5.6\ Ah = I_{start}t_{start} + I_{max}t_{max} + I_{idle}t_{idle} + I_{COM}t_{COM}$$

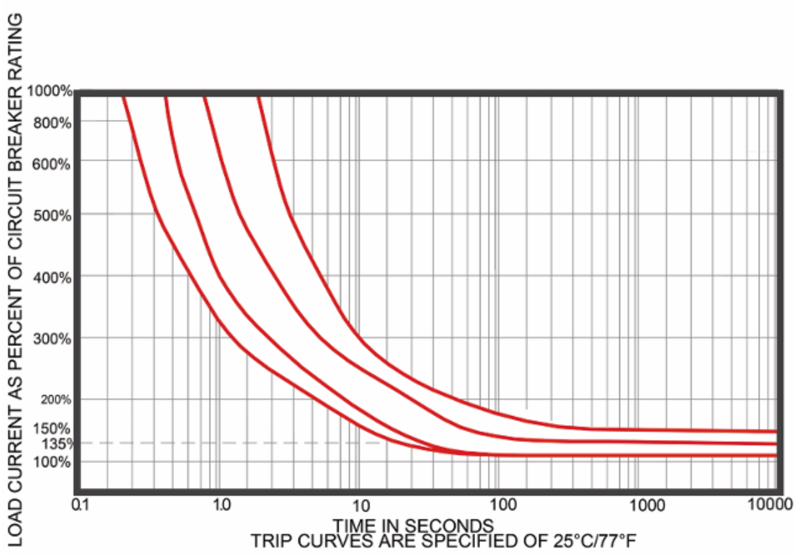
Available time in idle mode

$$t_{idle} = \frac{\left[5.6Ah - \left((2.6A) * \left(2\ sec * \frac{1\ hr}{3600\ sec}\right) + (10.18A) * \left(10\ min * \frac{1\ hr}{60\ min}\right) + (6.16A) * \left(3\ sec * \frac{1\ hr}{3600\ sec}\right)\right)\right]}{2.6A} = 2.293\ hr$$

Available total time

$$t_{total} = t_{idle} + t_{max} = 2.293\ hr + 10min * \frac{1\ hr}{60\ min} = 2.460\ hr$$

Circuit Breakers Trip Curves

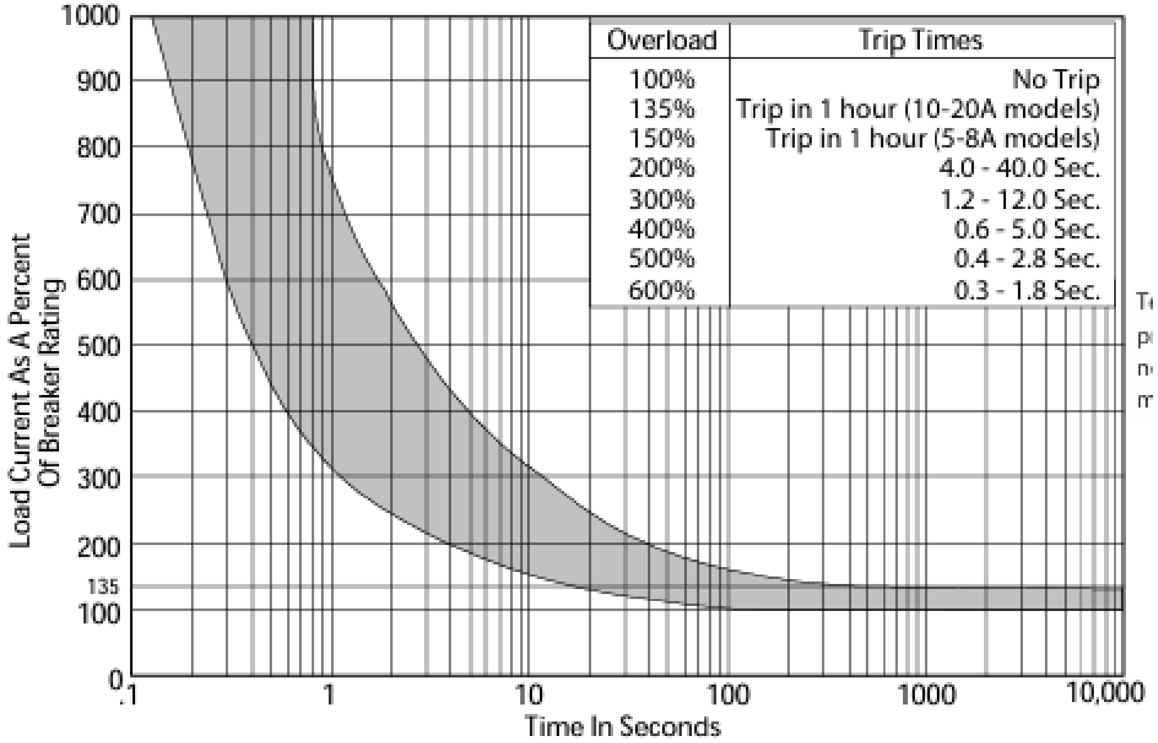


Overload	Trip Time sec	
	0.25~2.5A	3~20A
600%	0.53-3.1	0.34-1.05
500%	0.67-3.8	0.42-1.75
400%	1.0-5.5	0.62-3.05
300%	2.85-10.0	1.50-5.5
200%	7.5-52.0	5.2-30.0
150%	Trip in 1 Hour	-
135%	-	Trip in 1 Hour
100%	Hold, No Trip	

2.5 A

Time vs. Current Trip Curve @ +25°C

Overload	Trip Times
100%	No Trip
135%	Trip in 1 hour (10-20A models)
150%	Trip in 1 hour (5-8A models)
200%	4.0 - 40.0 Sec.
300%	1.2 - 12.0 Sec.
400%	0.6 - 5.0 Sec.
500%	0.4 - 2.8 Sec.
600%	0.3 - 1.8 Sec.

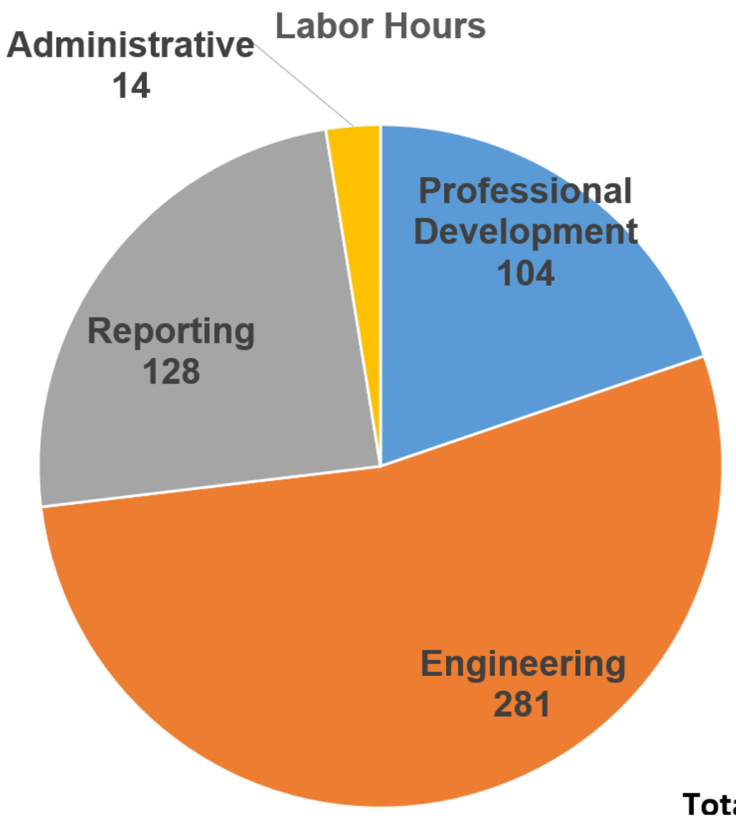


6 A, 10 A

Wiring Diagram

Hours

Team Hours by Category



Total 526.0

Team Hours by Member